



On finite difference methods for the solution of the Schrödinger equation

T.E. Simos^{a,*}, P.S. Williams^b

^a*Section of Mathematics, Department of Civil Engineering, School of Engineering, Democritus University of Thrace, GR 671 00, Xanthi, Greece.*

^b*Department of Computing, Information Systems and Mathematics, Faculty of Human Sciences, London Guildhall University, 100 Minories, London, EC3N 1JY, UK*

Received 21 September 1998; accepted 9 February 1999

Abstract

A review for the numerical methods used for the solution of the Schrödinger equation is presented. © 1999 Elsevier Science Ltd. All rights reserved.

Keywords: Schrödinger equation; Eigenvalue problems; Resonance problems; Scattering problems; Phase shift problems; Finite differences; Multistep methods; Exponential fitting; Bessel and Neumann fitting; Phase-lag coupled differential equations

1. Introduction

The radial Schrödinger equation has the form:

$$y''(x) + f(x)y(x) = 0 \quad (1)$$

where $0 \leq x < \infty$ and $f(x) = E - l(l+1)/x^2 - V(x)$. We call the term $l(l+1)/x^2$ the *centrifugal potential* and the function $V(x)$ the *potential*, where $V(x) \rightarrow 0$ as $x \rightarrow \infty$. According to the sign of the energy E there are two main categories of problems for the Eq. (1) (see for details Simos (1990a)). In Eq. (1), E is a real number denoting the *energy*, l is a given integer and V is a given function which denotes the *potential*. The function $W(x) = l(l+1)/x^2 + V(x)$ denotes the *effective potential*, which satisfies $W(x) \rightarrow 0$ as $x \rightarrow \infty$. The boundary conditions are:

$$y(0) = 0 \quad (2)$$

and a second boundary condition, for large values of x , determined by physical considerations.

There is a real need for the numerical solution of the one-dimensional Schrödinger equation in many scientific areas. Some of these areas are the nuclear physics, the physical chemistry, the theoretical physics and chemistry (Blatt, 1967; Herzberg, 1950).

There is much activity in the area of the solution of the radial Schrödinger equation (1). The most important characteristics of an efficient method for the solution of the problem (1) are the accuracy and the computational efficiency. The development of methods with the above mentioned characteristics is an open problem.

The purpose of this paper is to develop a review for the numerical methods for the solution of the Schrödinger equation. In Section 2 the basic theories for the construction of the methods are presented. In Sections 3–9 the methods are presented. Finally in Section 10 numerical results are presented and comments are made.

* Corresponding author. Tel.: +30-54129706; fax: +301-94-13-189.

E-mail address: tsimos@mail.ariadne-t.gr (T.E. Simos)

2. Theory for constructing methods for the numerical solution of the Schrödinger equation

2.1. Phase-lag analysis for two-step methods

We investigate the numerical integration of the problem:

$$y'' = f(x, y), \quad y(x_0) = y_0, \quad y'(x_0) = y'_0 \quad (3)$$

To examine the stability properties of methods for solving the initial-value problem (3) Lambert and Watson (1976) introduced the scalar test equation

$$y'' = -w^2 y \quad (4)$$

and its interval of periodicity. When we apply a symmetric two-step method to the scalar test (4), we obtain a difference equation of the form

$$y_{n+1} - 2Q(s)y_n + y_{n-1} = 0 \quad (5)$$

where $s = wh$, h is the step length, $Q(s) = \frac{B(s)}{A(s)}$ where $B(s)$ and $A(s)$ are polynomials in s and y_n is the computed approximation to $y(nh)$, $n = 0, 1, 2, \dots$. For explicit methods $A(s) = 1$.

The characteristic equation associated with Eq. (5) is,

$$z^2 - 2Q(s)z + 1 = 0 \quad (6)$$

We have the following definitions:

Definition 1. (Thomas, 1984) The method (5) with characteristic equation (6) is unconditionally stable if $|z_1| \leq 1$ and $|z_2| \leq 1$ for all values of wh , where z_1 and z_2 are the roots of Eq. (6).

Definition 2. Following Lambert and Watson (1976) we say that the numerical method (5) has an interval of periodicity $(0, H_0^2)$. if, for all $s^2 \in (0, H_0^2)$, z_1 and z_2 satisfy:

$$z_1 = e^{i\theta(s)} \quad \text{and} \quad z_2 = e^{-i\theta(s)}, \quad (7)$$

where $\theta(s)$ is a real function of s .

Definition 3. (Lambert and Watson, 1976) The method (5) is P -stable if its interval of periodicity is $(0, \infty)$.

Using these definitions, we have the following theorems (for the proofs see Simos and Toulgelidis (1996a))

Theorem 1. A method which has the characteristic equation (6), has an interval of periodicity $(0, H_0^2)$, if for all $s^2 \in (0, H_0^2)$, $|Q(s)| < 1$.

Theorem 2. For a method which has an interval of periodicity $(0, H_0^2)$ we can write:

$$\cos[\theta(s)] = Q(s), \quad \text{where } s^2 \in (0, H_0^2). \quad (8)$$

Definition 4. (van der Houwen and Sommeijer, 1987). For any method corresponding to the characteristic (6), the quantity:

$$t = s - \cos^{-1}[Q(s)] \quad (9)$$

is called the dispersion or the phase error or the phase-lag of the method. If $t = O(s^{q+1})$ as $s \rightarrow 0$ the order of phase-lag is q .

From Definition 4 Coleman (1989) makes the following remark:

Remark 1. If the order of dispersion is $2r$ then:

$$\begin{aligned} t &= cs^{2r+1} + O(s^{2r+3}) \Rightarrow \cos(s) - Q(s) \\ &= \cos(s) - \cos(s-t) = cs^{2r+2} + O(s^{2r+4}), \end{aligned} \quad (10)$$

where t is the phase-lag of the method.

2.2. Phase-lag analysis of general symmetric $2k$ -step, $k \in N$ methods

In recent years there has been considerable interest in the numerical solution of second order periodic initial-value problems (Lambert and Watson, 1976; Thomas, 1984; van der Houwen and Sommeijer, 1987; Simos, 1993a; Coleman, 1989; Simos, 1990a)

$$y'' = f(x, y), \quad y(x_0) = y_0, \quad y'(x_0) = y'_0 \quad (11)$$

To investigate the absolute stability properties, we introduce the scalar test equation:

$$y'' = -w^2 y \quad (12)$$

When we apply a symmetric $2k$ method to the scalar test (12), we obtain a difference equation of the form

$$\begin{aligned} A_k(H)y_{n+k} + \dots + A_1(H)y_{n+1} + A_0(H)y_n \\ + A_1(H)y_{n-1} \\ + \dots + A_k(H)y_{n-k} = 0 \end{aligned} \quad (13)$$

where $H = wh$, h is the step length and $A_0(H), A_1(H), \dots, A_k(H)$ are polynomials of H and y_n is the computed approximation to $y(nh)$, $n = 0, 1, 2, \dots$

The characteristic equation associated with Eq. (13) is

$$A_k(H)s^k + \dots + A_1(H)s + A_0(H) + A_1(H)s^{-1} + \dots + A_k(H)s^{-k} = 0 \quad (14)$$

From Lambert and Watson (1976) we have the following definitions:

Definition 5. A symmetric $2k$ -step method with the characteristic equation given by Eq. (14) is said to have an interval of periodicity (H_0^2, H_1^2) if, for all $H \in (H_0^2, H_1^2)$, the roots $s_i, i = 1, \dots, 2k$ of Eq. (14) satisfy:

$$s_1 = e^{i\theta(H)}, \quad (15)$$

$$s_2 = e^{-i\theta(H)}, \quad \text{and} \quad |s_i| \leq 1, \quad i = 3, \dots, 2k$$

where $\theta(H)$ is a real function of H .

Definition 6. For any method corresponding to the characteristic equation (14) the phase-lag is defined as the leading term in the expansion of

$$t = H - \theta(H) \quad (16)$$

Then if the quantity $t = O(H^{q+1})$ as $H \rightarrow 0$, the order of phase-lag is q .

The phase-lag theory is also valid for the case where w is imaginary. In that case, we now have an exponential error which corresponds to the phase-lag.

Theorem 3. For all H in the interval of periodicity, the symmetric $2k$ -step method with characteristic equation given by Eq. (14) has phase-lag order q and phase-lag constant c given by

$$-cH^{q+2} + O(H^{q+3}) = \frac{2A_k(H)\cos(kH) + \dots + 2A_j(H)\cos(jH) + \dots + A_0(H)}{2k^2A_k(H) + \dots + 2j^2A_j(H) + \dots + 2A_1(H)} \quad (17)$$

Proof. If we put $s = e^{i\theta(H)}$, then Eq. (14) becomes

$$2A_k(H)\cos(k\theta(H)) + \dots + 2A_j(H)\cos(j\theta(H)) + \dots + A_0(H) = 0 \quad (18)$$

By definition the phase-lag order q and phase-lag constant c are given by

$$t = H - \cos(\theta(H)) = cH^{q+1} + O(H^{q+2}) \quad (19)$$

then since

$$\theta(H) = H - t \quad (20)$$

we may show from trigonometric expansions that

$$\cos(\theta(H)) = \cos(H - t) = \cos H + cH^{q+2} + O(H^{q+3}) \quad (21)$$

$$\sin(\theta(H)) = \sin(H - t) = \sin H - cH^{q+1} + O(H^{q+2}) \quad (22)$$

By an inductive argument using the familiar identities

$$\cos(j\theta(H)) = \cos((j-1)\theta(H))\cos(\theta(H)) - \sin((j-1)\theta(H))\sin(\theta(H)) \quad (23)$$

$$\sin(j\theta(H)) = \sin((j-1)\theta(H))\cos(\theta(H)) + \cos((j-1)\theta(H))\sin(\theta(H)) \quad (24)$$

it is now straightforward to show that

$$\cos(j\theta(H)) = \cos(jH) + c_j^2 H^{q+2} + O(H^{q+3}) \quad (25)$$

$$\sin(j\theta(H)) = \sin(jH) - c_j H^{q+1} + O(H^{q+2}) \quad (26)$$

Then substituting Eqs. (25) and (26) into (18) for $j = 1, 2, \dots, k$ will give the result.

The converse of the theorem may also be easily shown. The converse states that if Eq. (17) is true then the method will have phase-lag order q and phase-lag constant c . To prove this, we suppose that the method has phase-lag order p and phase-lag constant d . This means

$$t = H - \theta(H) = dH^{p+1} + O(H^{p+1}) \quad (27)$$

Then we may show by trigonometric expansions that

$$\cos(j\theta(H)) = \cos(jH) + dj^2 H^{p+2} + O(H^{p+3}) \quad (28)$$

Substituting for $\cos(jH)$ in Eq. (17) and using Eq. (18) we have

$$-2dH^{p+1} \sum_{j=0}^k A_j j^2 = -2cH^{q+1} \quad (29)$$

$$\sum_{j=0}^k A_j j^2 + O(H^{q+2})$$

Equating highest powers gives

$$p = q \quad \text{and} \quad d = c \quad (30)$$

Table 1
The m -stage explicit Runge–Kutta method

0				
a_1	b_{10}			
a_2	b_{20}	b_{21}		
$\vdots$	$\vdots$	$\vdots$		
a_m	$b_{m,0}$	$b_{m,1}$	$\dots$	$b_{m,m-1}$
	c_0	c_1	$\dots$	c_{m-1}
				c_m

i.e., the theorem is proved.

The formula proposed from the above theorem gives us a direct method to calculate the phase-lag of any symmetric $2k$ -step method. For the symmetric four-step methods the phase-lag order and the phase-lag constant are given by Eq. (17) with $k = 2$.

The calculation of the exponential error follows the same principle. Now w is imaginary and $\cos$ is replaced by $\cosh$ in formula (17).

2.3. Phase-lag of the Runge–Kutta methods

To develop the basic theory on the calculation of the phase-lag of a Runge–Kutta method we use the test equation,

$$y' = ivy, \quad v \text{ is real.} \quad (31)$$

Based on the reasons fully described by van der Houwen and Sommeijer (1987) we shall confine our considerations to homogeneous phase-lag and based on its definition given in (van der Houwen and Sommeijer, 1987), we shall use a test equation with an exact solution of the form e^{ivx} . However, as it is shown by numerical results, inhomogeneous problems can be successfully dealt with by increasing the order of homogeneous phase-lag. By comparing the exact and the numerical solution for this equation, and by requiring that these solutions are in phase with maximal order in the step-size h , we derive the so-called *phase-lag relation*.

For first-order equations we write the m -stage explicit Runge–Kutta method in the matrix form given in Table 1.

Application of the above method to Eq. (31) yields the numerical solution:

$$y_n = a_*^n v_0 \quad \text{and} \quad (32)$$

$$a_* = A_m(H^2) + iHB_m(H^2), \quad H = vh,$$

where

$$A_m(H^2) = 1 - t_2H^2 + t_4H^4 + t_6H^6 \dots,$$

$$B_m(H^2) = 1 - t_3H^2 + t_5H^4 + t_7H^6 \dots, \quad (33)$$

are polynomials in H^2 , completely defined by Runge–Kutta parameters a_i, b_{ij} and c_i , $i = 1, \dots, m$, $j = 1, \dots, i-1$, $l = 0, \dots, m$. The dissipative factor is $a_* = a_*(H)$, and y_n denotes the approximation to $y(x_n)$, where $x_n = nh$, $n = 0, 1, \dots$

A comparison of Eq. (32) with the solution of (31), leads to the following definition of the *dispersion or phase error or phase-lag and the dissipative error*.

Definition 7. (van der Houwen and Sommeijer, 1987; Simos, 1993a). In the explicit m -stage Runge–Kutta method, presented in Table 1, the quantities:

$$t(H) = H - \arg[a_*(H)], \quad a(H) = 1 - |a_*(H)|. \quad (34)$$

are respectively called the *phase-lag and the dissipative error*. If $t(H) = O(H^{r+1})$ and $a(H) = O(H^{s+1})$ then the method is said to be of *phase-lag order r and dissipative order s* .

Remark 2. (van der Houwen and Sommeijer, 1987; Simos 1993a). From the definition (34) it follows that,

$$a(H) = 1 - \sqrt{[A_m^2(H^2) + H^2 B_m^2(H^2)]}. \quad (35)$$

The interval $(0, H)$ for which $a(H) \geq 0$ is called *interval of imaginary stability*.

Also we have the following theorem.

Theorem 4. For the Runge–Kutta method given by Table 1 and Eq. (31) we have the following formula for the direct calculation of the phase-lag order r and the phase-lag constant c

$$\tan(H) - H \left[\frac{B_m(H^2)}{A_m(H^2)} \right] = cH^{r+1} + O(H^{r+3}) \quad (36)$$

Proof. From the definition, we have that

$$t(H) = H - \arctan \left[H \frac{B_m(H^2)}{A_m(H^2)} \right]. \quad (37)$$

So it follows that

$$H - \arctan \left[H \frac{B_m(H^2)}{A_m(H^2)} \right] = cH^{r+1} + O(H^{r+3})$$

$$\arctan \left[H \frac{B_m(H^2)}{A_m(H^2)} \right] = H - cH^{r+1} + O(H^{r+3})$$

$$\tan\left(\arctan\left[H\frac{B_m(H^2)}{A_m(H^2)}\right]\right) = \tan[H - cH^{r+1} + O(H^{r+3})]$$

$$\left[H\frac{B_m(H^2)}{A_m(H^2)}\right] = \tan(H) - cH^{r+1} + O(H^{r+3}) \quad (38)$$

and hence Eq. (36) follows.

2.4. Exponential multistep methods

For the numerical solution of the initial value problem

$$y^{(r)} = f(x, y), \quad y^{(j)}(A) = 0, \quad j = 0, 1, \dots, r-1 \quad (39)$$

multistep methods of the form

$$\sum_{i=0}^k a_i y_{n+i} = h^r \sum_{i=0}^k b_i f(x_{n+i}, y_{n+i}) \quad (40)$$

can be used over the equally spaced intervals $\{x_i\}_{i=0}^k \in [A, B]$ and $h = |x_{i+1} - x_i|, i = 0, \dots, k-1$.

The method (40) is associated with the operator

$$L(x) = \sum_{i=0}^k [a_i z(x + ih) - h^r b_i z^{(r)}(x + ih)] \quad (41)$$

where z is a continuously differentiable function.

Definition 8. The multistep method (40) is called algebraic (or exponential) of order p if the associated linear operator L vanishes for any linear combination of the linearly independent functions $1, x, x^2, \dots, x^{p+r-1}$ (or $\exp(v_0, x), \exp(v_1, x), \dots, \exp(v_{p+r-1}, x)$ where $v_i, i = 0, 1, \dots, p+r-1$ are real or complex numbers).

Remark 3. (Lyche, 1972; Simos, 1990a). If $v_i = v$ for $i = 0, 1, \dots, n, n \leq p+r-1$ then the operator L vanishes for any linear combination of $\exp(vx), x \exp(vx), x^2 \exp(vx), \dots, x^n \exp(vx), \exp(v_{n+1}x), \dots, \exp(v_{p+r-1}, x)$.

Remark 4. (Lyche, 1972; Simos, 1990a). Every exponential multistep method of order p corresponds in a unique way, to an algebraic multistep method of order p (by setting $v_i = 0$, for all i).

Lemma 1. (For proof see (Lyche, 1972; Henrici, 1962). Consider an operator L of the form (41). With $v \in \mathbb{C}$, $h \in \mathbb{R}, n \geq r$ if $v = 0$, and $n \geq 1$ otherwise, then we have

$$L[x^m \exp(vx)] = 0, \quad m = 0, 1, \dots, n-1, \quad (42)$$

$$L[x^n \exp(vx)] \neq 0$$

if and only if the function ϕ has a zero of exact multiplicity s at $\exp(vh)$, where $s = n$ if $v \neq 0$, and $s = n-r$ if $v = 0$, $\phi(w) = \rho(w)/\log^r w - \sigma(w)$, $\rho(w) = \sum_{i=0}^k a_i w^i$ and $\sigma(w) = \sum_{i=0}^k b_i w^i$.

Proposition 1. (For proof see (Raptis, 1984; Simos, 1990a). Consider an operator L with

$$L[\exp(\pm v_i x)] = 0, j = 0, 1, \dots, k \leq \frac{p+r-1}{2} \quad (43)$$

then for given a_i and p with $a_i = (-1)^i a_{k-i}$ there is a unique set of b_i such that $b_i = b_{k-i}$.

2.5. The derivation of exponentially fitted methods for general problems

Consider the construction of an exponentially fitted multistep method (40) which exactly integrates the set of functions $\{\exp(\pm v_j x)\}_{j=0}^k$. We will use this for the numerical solution of the general problem (39).

From Lemma 1, we obtain the equations

$$\rho[\exp(\pm v_j h)] - (\pm v_j h)^r \sigma[\exp(\pm v_j h)] = 0 \quad (44)$$

or equivalently

$$\sum_{i=0}^k [a_i \exp(\pm v_j h) - (\pm v_j h)^r b_i \exp(\pm v_j h)] = 0, \quad (45)$$

$$j = 0, 1, \dots, n$$

where $n \leq k$ and $a_i, b_i, i = 0, \dots, k$ are the coefficients of the multistep method (40).

We investigate here the case where k is a positive number. Then, from Proposition 1, we have a set of k equations:

$$a_i = (-1)^i a_{k-i}, \quad b_i = b_{k-i}, \quad i = 0, 1, \dots, k. \quad (46)$$

We let $a_k = 1$, which is the convention adopted for all families of known multistep methods. Then Eqs. (45) and (46) give the following system of equations

$$2 \sum_{i=1}^{k/2-1} a_i \sinh\left[\left(\frac{k}{2} - i\right)w_j\right] + a_{k/2}$$

$$- w_j^r \left[2 \sum_{i=0}^{k/2-1} b_i \cosh\left[\left(\frac{k}{2} - i\right)w_j\right] + b_{\frac{k}{2}} \right] \quad (47)$$

$$= -2 \sinh\left(\frac{k w_j}{2}\right),$$

for $r = 1, 3, 5, \dots$

$$\begin{aligned}
& 2 \sum_{i=1}^{k/2-1} a_i \cosh \left[\left(\frac{k}{2} - i \right) w_j \right] + a_{k/2} \\
& - w_j^r \left[2 \sum_{i=0}^{k/2-1} b_i \cosh \left[\left(\frac{k}{2} - i \right) w_j \right] + b_{k/2} \right] \\
& = -2 \cosh \left(\frac{k w_j}{2} \right),
\end{aligned} \quad (48)$$

for $r = 2, 4, 6, \dots$

where $w_j = v_j h$ and $j = 0, 1, \dots, k$. We now prove that the system of equations either (i) has a unique solution when $w_i \neq \pm w_j$, or (ii) lead to so called ‘indeterminate’ expressions of the form $\left(\frac{0}{0}\right)$ when $w_i = \pm w_j$ for some i and j .

Let $X(w)$ and $Y(w)$ ($w = v h$) be the matrices of the unknown coefficients in the systems of Eqs. (47) and (48), respectively. Consider case (i). In order to make the matrices $X(w)$ (or $Y(w)$) singular then their columns would have to be linearly dependent. The elements in a row consist of terms like $\cosh M w_j$, $\sinh N w_j$ and powers of w_j . The multiple angle hyperbolic functions can be expressed in terms of powers of $\cosh w_j$, $\sinh w_j$ and their products. These with powers of w_j form a linearly independent set of functions. Therefore the columns cannot be linearly dependent. Hence in this case $\det X(w) \neq 0$ (or $\det Y(w) \neq 0$). Thus the system of Eqs. (47) and (48) has a unique solution.

Consider case (ii). Here we simply have two rows of the matrix of coefficients the same and hence $\det X(w) = 0$ (or $\det Y(w) = 0$). Similarly we have the right-hand side of two of the equations in (10) or (11) the same so that the numerator determinant which is formed when a column of $X(w)$ (or $Y(w)$) is replaced by the right hand column will also give two identical rows. Hence the numerator determinant is 0. In these cases L'Hospital's rule must be used to evaluate these expressions.

In the present paper we investigate the case $r = 2$.

3. Exponentially fitted methods

A fruitful way for developing efficient methods for the solution of the Schrödinger equation is to use exponential fitting. Raptis and Allison (1978) have derived a Numerov type exponentially fitted method. The computational results obtained in their paper indicate that these fitted methods are much more efficient than Numerov's method for the solution of the Schrödinger equation. Since this paper, exponential fitting has been the subject of great activity which will be discuss later. An interesting paper in this general area is that of Ixaru and Rizea (1980). They showed that for stiff

oscillating problems of the Schrödinger equation, it is generally more efficient to derive methods which exactly integrate functions of the form

$$\begin{aligned}
& \{1, x, x^2, \dots, x^l, \exp(\pm vx), x \exp(\pm vx) \\
& \dots, x^m \exp(\pm vx)\}.
\end{aligned} \quad (49)$$

where v is the frequency of the problem, than to use classical exponential fitting methods. The reason for this is explained in Simos (1990a). In the following we categorise all the exponentially fitted methods and give the basic properties for these methods. The coefficients in each of these methods are given in the specified references.

3.1. Linear multistep methods

3.1.1. Two-step methods

The exponentially fitted methods of this category are of the form

$$y_{n+1} + a_1 y_n + y_{n-1} = h^2 \left(b_0 (y''_{n+1} + y''_{n-1}) + b_1 y''_n \right) \quad (50)$$

Using Eq. (50) a group of exponentially fitted methods has been produced. These are found in:

1. Raptis and Allison (1978). This integrates exactly:

$$\{1, x, x^2, x^3, \exp(\pm vx)\} \quad (51)$$

2. Ixaru and Rizea (1980). This integrates exactly:

$$\{1, x, \exp(\pm vx), x \exp(\pm vx)\} \quad (52)$$

3. Raptis, (1982). This integrates exactly:

$$\{\exp(\pm vx), x \exp(\pm vx), x^2 \exp(\pm vx)\} \quad (53)$$

It is instructive to note that Numerov's method integrates exactly $\{1, x, x^2, x^3, x^4, x^5\}$.

3.1.2. Four-step methods

One family of the exponentially fitted methods of this category are of the form

$$\begin{aligned}
& y_{n+2} + a_1 y_{n+1} + 2y_n + a_1 y_{n-1} + y_{n-2} \\
& = h^2 \left(b_0 (y''_{n+2} + y''_{n-2}) \right. \\
& \quad \left. + b_1 (y''_{n+1} + y''_{n-1}) + b_2 y''_n \right)
\end{aligned} \quad (54)$$

Using Eq. (54) a group of exponentially fitted methods has been produced:

1. Raptis (1983). This integrates exactly:

$$\{1, x, x^2, x^3, x^4, x^5, \exp(\pm wx)\} \quad (55)$$

2. Raptis (1981). This integrates exactly:

$$\{1, x, \exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx)\} \quad (56)$$

3. Simos (1990b). This integrates exactly:

$$\{\exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx), x^3 \exp(\pm wx)\} \quad (57)$$

Another family of the exponentially fitted methods of this category are of the form

$$y_{n+2} + a_0 y_{n+1} + a_0 y_{n-1} + y_{n-2} = h^2 \left(b_0 (y''_{n+2} + y''_{n-2}) + b_1 (y''_{n+1} + y''_{n-1}) + b_2 y''_n \right) \quad (58)$$

Using Eq. (58) a group of exponentially fitted methods is produced:

3.1. Simos (1991a). This integrates exactly:

$$\{1, x, x^2, x^3, x^4, x^5, \exp(\pm wx)\} \quad (59)$$

3.2. Simos (1991a). This integrates exactly:

$$\{1, x, x^2, x^3, \exp(\pm wx), x \exp(\pm wx)\} \quad (60)$$

3.3. Simos (1991a). This integrates exactly:

$$\{1, x, \exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx)\} \quad (61)$$

3.4. Simos (1991a). This integrates exactly:

$$\{\exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx), x^3 \exp(\pm wx)\} \quad (62)$$

3.2. Hybrid sixth algebraic order methods

The exponentially fitted methods of this category are of the form

$$y_{n+1} + a_1 y_n + y_{n-1} = h^2 \left(b_0 (y''_{n+1} + y''_{n-1}) + b_1 \left(\hat{y}''_{n+1} \frac{1}{n+2} + \hat{y}''_{n-1} \frac{1}{n-2} \right) + b_2 y''_n \right) \quad (63)$$

where

$$\hat{y}''_{n+1} \frac{1}{n+2} = A_1 (y_{n+1} + y_{n-1}) + A_2 y_n + h^2 [A_3 (y''_{n+1} + y''_{n-1}) + A_4 y''_n] \quad (64)$$

$$\hat{y}''_{n-1} \frac{1}{n-2} = B_1 (y_{n+1} + y_{n-2}) + B_2 (y_n + y_{n-1}) + h^2 [B_3 (y''_{n+1} + y''_{n-2}) + B_4 (y''_n + y''_{n-1})] \quad (65)$$

Using Eqs. (64) and (65) another group of exponentially fitted methods has been produced:

1. Raptis and Cash (1987). This integrates exactly:

$$\{1, x, x^2, x^3, x^4, x^5, \exp(\pm wx)\} \quad (66)$$

2. Cash et al. (1990). This integrates exactly:

$$\{1, x, x^2, x^3, \exp(\pm wx), x \exp(\pm wx)\} \quad (67)$$

3. Simos (1992a). This integrates exactly:

$$\{1, x, \exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx)\} \quad (68)$$

4. Simos (1996a). This integrates exactly:

$$\{\exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx), x^3 \exp(\pm wx)\} \quad (69)$$

Error analysis for most of these methods are given by Simos (1993b).

3.3. Predictor–corrector methods

3.3.1. Implicit fourth-order methods

Simos (1997a) has considered the following method

$$\hat{y}_n = y_n - ah^2 (y''_{n+1} - 2y''_n + y''_{n-1})$$

$$y_{n+1} + a_1 y_n + y_{n-1} = h^2 [b_0 (y''_{n+1} + y''_{n-1}) + b_1 \hat{y}''_n] \quad (70)$$

Using Eq. (70) an exponentially fitted method has been produced (see Simos, 1997a) which integrates exactly:

$$\{\exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx), x^3 \exp(\pm wx)\} \quad (71)$$

Thomas et al., (1996) have considered the following family of methods

$$\hat{y}_n = y_n - ah^2(y''_{n+1} - 2y''_n + y''_{n-1})$$

$$\hat{\hat{y}}_n = y_n - bh^2(y''_{n+1} - 2\hat{y}''_n + y''_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = h^2 \left[b_0(y''_{n+1} + y''_{n-1})b_1\hat{y}''_n \right] \quad (72)$$

Using Eq. (72) the another group of exponentially fitted methods has been produced (Thomas et al., 1996). These integrate exactly the following functions:

$$\text{Case I} \quad \{1, x, x^2, x^3, x^4, x^5, x^6, x^7, \exp(\pm wx)\} \quad (73)$$

$$\text{Case II} \quad \{1, x, x^2, x^3, x^4, x^5, \exp(\pm wx), x \exp(\pm wx)\} \quad (74)$$

$$\text{Case III} \quad \{1, x, x^2, x^3, \exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx)\} \quad (75)$$

$$\text{Case IV} \quad \{1, x, \exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx), x^3 \exp(\pm wx)\} \quad (76)$$

Simos (1997b) has derived the final exponentially fitted method in this group which integrates exactly:

$$\text{Case V} \quad \{\exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx), x^3 \exp(\pm wx), x^4 \exp(\pm wx)\} \quad (77)$$

Simos and Williams (1997a) have considered the following family of methods

$$\hat{y}_n = y_n - ah^2(y''_{n+1} - 2y''_n + y''_{n-1})$$

$$\hat{\hat{y}}_n = y_n - bh^2(y''_{n+1} - 2\hat{y}''_n + y''_{n-1})$$

$$\hat{\hat{\hat{y}}}_n = y_n - ch^2(y''_{n+1} - 2\hat{\hat{y}}''_n + y''_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = h^2 \left[b_0(y''_{n+1} + y''_{n-1}) + \hat{\hat{y}}''_n \right] \quad (78)$$

Using Eq. (78) a group of exponentially fitted methods has been produced (Simos and Williams, 1997a). These integrate exactly:

$$\text{Case I} \quad \{1, x, x^2, x^3, x^4, x^5, x^6, x^7, x^8, x^9, \exp(\pm wx)\} \quad (79)$$

$$\text{Case II} \quad \{1, x, x^2, x^3, x^4, x^5, x^6, x^7, \exp(\pm wx), x \exp(\pm wx)\} \quad (80)$$

$$\text{Case III} \quad \{1, x, x^2, x^3, x^4, x^5, \exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx)\} \quad (81)$$

$$\text{Case IV} \quad \{1, x, x^2, x^3, \exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx), x^3 \exp(\pm wx)\} \quad (82)$$

$$\text{Case V} \quad \{1, x, \exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx), x^3 \exp(\pm wx), x^4 \exp(\pm wx)\} \quad (83)$$

$$\text{Case VI} \quad \{\exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx), x^3 \exp(\pm wx), x^4 \exp(\pm wx), x^5 \exp(\pm wx)\} \quad (84)$$

3.4. Explicit fourth-order methods

Simos (1997c) has derived the following family of explicit exponentially fitted method

$$\hat{y}_{n+1} = 2y_n - y_{n-1} + h^2 y''_n$$

$$\hat{y}_n = y_n - ah^2(\hat{y}''_{n+1} - 2y''_n + y''_{n-1})$$

$$y_{n+1} + a_1 y_n + y_{n-1} = h^2 \left[b_0(\hat{y}''_{n+1} + y''_{n-1}) + b_1 y''_n \right] \quad (85)$$

Using Eq. (85) an exponentially fitted method has been produced (Simos, 1997) which integrates exactly:

$$\{\exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx), x^3 \exp(\pm wx)\}. \quad (86)$$

Table 2

Properties of the two-step fourth algebraic order exponentially fitted methods ($S = \{H^2: H = sq\pi, q = 1, 2, \dots\}$). The quantities m and p are defined by Eq. (49). A.O. is the algebraic order of the method. Int. Per. is the interval of periodicity of the method. I = Implicit, E = Explicit

Method	A.O.	I/E	Interval of periodicity	Integrated exponential functions
Numerov's method	4	I	(0, 6)	$1, x, x^2, x^3, x^4, x^5$
Derived by Raptis and Allison (1978)	4	I	$(0, \infty) - S$	$m = 0, p = 3$
Derived by Ixaru and Rizea (1980)	4	I	$(0, \infty) - S$	$m = 1, p = 1$
Derived by Raptis (1982)	4	I	$(0, \infty) - S$	$m = 2, p = 0$
Method of Simos (1997a)	4	I	$(0, \infty) - S$	$m = 3, p = 0$
Method of TMS—Case I ^a	4	I	$(0, \infty) - S$	$m = 0, p = 7$
Method of TMS—Case II ^a	4	I	$(0, \infty) - S$	$m = 1, p = 5$
Method of TMS—Case III ^a	4	I	$(0, \infty) - S$	$m = 2, p = 3$
Method of TMS—Case IV ^a	4	I	$(0, \infty) - S$	$m = 3, p = 1$
Method of Simos (1997b)	4	I	$(0, \infty) - S$	$m = 4, p = 0$
Method of SiWi—Case I ^b	4	I	$(0, \infty) - S$	$m = 0, p = 9$
Method of SiWi—Case II ^b	4	I	$(0, \infty) - S$	$m = 1, p = 7$
Method of SiWi—Case III ^b	4	I	$(0, \infty) - S$	$m = 2, p = 5$
Method of SiWi—Case IV ^b	4	I	$(0, \infty) - S$	$m = 3, p = 3$
Method of SiWi—Case V ^b	4	I	$(0, \infty) - S$	$m = 4, p = 1$
Method of SiWi—Case VI ^b	4	I	$(0, \infty) - S$	$m = 5, p = 0$
Method of Simos (1997c)	4	E	$(0, \infty) - S$	$m = 3, p = 0$

^a TMS = Thomas Mitsou and Simos TMS = (Thomas et al., 1996).

^b SiWi = (Simos and Williams, 1997a).

In Table 2 we present the basic properties of the fourth algebraic order two-step methods.

3.5. Sixth-order methods

3.5.1. Four-step predictor–corrector methods

Simos and Mitsou (1994) have considered the following family of methods

$$\hat{y}_n = y_n - ah^2(y''_{n+2} - 4y''_{n+1} + 6y''_n - 4y''_{n-1} + y''_{n-2})$$

$$y_{n+2} - 2y_{n+1} + 2y_n - 2y_{n-1} + y_{n-2}$$

$$= h^2[b_0(y''_{n+2} + y''_{n-2}) + b_1(y''_{n+1} + y''_{n-1}) + b_2\hat{y}_n] \quad (87)$$

Using Eq. (87) a group of following exponentially fitted methods have been produced by Simos and Mitsou (1994) which integrate exactly:

$$\text{Case I} \quad \{1, x, x^2, x^3, x^4, x^5, x^6, x^7, \exp(\pm wx)\} \quad (88)$$

$$\text{Case II} \quad \{1, x, x^2, x^3, x^4, x^5, \exp(\pm wx), x \exp(\pm wx)\} \quad (89)$$

$$\text{Case III} \quad \{1, x, x^2, x^3, \exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx)\} \quad (90)$$

Simos (1995a) has considered the following family of four-step predictor–corrector methods

$$\hat{y}_n = y_n - ah^2(y''_{n+2} - 4y''_{n+1} + 6y''_n - 4y''_{n-1} + y''_{n-2})$$

$$y_{n+2} - y_{n+1} - y_{n-1} + y_{n-2}$$

$$= h^2[b_0(y''_{n+2} + y''_{n-2}) + b_1(y''_{n+1} + y''_{n-1}) + b_2\hat{y}_n] \quad (91)$$

Using Eq. (91) another group of exponentially fitted methods have been produced by Simos and Mitsou, (1994) which integrate exactly:

$$\text{Case I} \quad \{1, x, x^2, x^3, x^4, x^5, x^6, x^7, \exp(\pm wx)\} \quad (92)$$

$$\text{Case II} \quad \{1, x, x^2, x^3, x^4, x^5, \exp(\pm wx), x \exp(\pm wx)\} \quad (93)$$

$$\text{Case III} \quad \{1, x, x^2, x^3, \exp(\pm wx), x \exp(\pm wx), x^2 \exp(\pm wx)\} \quad (94)$$

Simos (1998) has also considered the following four-step predictor–corrector method

Table 3

Properties of some two-step exponentially fitted methods ($S = \{H^2: H = sq\pi, q = 1, 2, \dots\}$). The quantities m and p are defined by Eq. (49). A.O. is the algebraic order of the method. Int. Per. is the interval of periodicity of the method. N.O.S. is the number of steps of the method. I.E.F. = Integrated exponential functions)

Method	A.O.	N.O.S	Int. Per.	I.E.F
Derived by Raptis and Cash (1987)	6	2	$(0, \infty) - S$	$m = 0, p = 5$
Derived by Cash et al., (1990)	6	2	$(0, \infty) - S$	$m = 1, p = 3$
Derived by Simos (1992a)	6	2	$(0, \infty) - S$	$m = 2, p = 0$
Method of SiMi—Case I ^a	6	4	$(0, \infty) - S$	$m = 0, p = 7$
Method of SiMi—Case II ^a	6	4	$(0, \infty) - S$	$m = 1, p = 5$
Method of SiMi—Case III ^a	6	4	$(0, \infty) - S$	$m = 2, p = 3$
Method of Simos—Case I (Simos, 1995a) ^a	6	4	$(0, \infty) - S$	$m = 0, p = 7$
Method of Simos—Case II (Simos, 1995a) ^a	6	4	$(0, \infty) - S$	$m = 1, p = 5$
Method of Simos—Case III (Simos, 1995a)	6	4	$(0, \infty) - S$	$m = 2, p = 3$
Method of Simos (Simos, 1998)	6	4	$(0, \infty) - S$	$m = 4, p = 0$
Method of TS—Case I ^b	6	2	$(0, \infty) - S$	$m = 0, p = 9$
Method of TS—Case II ^b	6	2	$(0, \infty) - S$	$m = 1, p = 7$
Method of TS—Case III ^b	6	2	$(0, \infty) - S$	$m = 2, p = 5$
Method of TS—Case IV ^b	6	2	$(0, \infty) - S$	$m = 3, p = 3$
Method of TS—Case V ^b	6	2	$(0, \infty) - S$	$m = 4, p = 1$

^a SiMi = (Simos and Mitsou, 1994).

^b TS = (Thomas and Simos, 1997).

$$\begin{aligned} \hat{y}_n &= y_n - ah^2(y''_{n+2} - 4y''_{n+1} + 6y''_n - 4y''_{n-1} + y''_{n-2}) \\ y_{n+1} &+ a_1y_n + y_{n-1} \\ &= h^2 \left[b_0(y''_{n+2} + y''_{n-2}) + b_1(y''_{n+1} + y''_{n-1}) \right. \\ &\quad \left. + b_2\hat{y}_n'' \right] \end{aligned} \quad (95)$$

Using Eq. (95) another exponentially fitted method has been produced by Simos (1998) which integrates exactly:

$$\begin{aligned} &\{\exp(\pm wx), x\exp(\pm wx), \\ &x^2\exp(\pm wx), x^3\exp(\pm wx), x^4\exp(\pm wx)\} \end{aligned} \quad (96)$$

3.5.2. Hybrid methods

Thomas and Simos (1997) have considered the following hybrid predictor–corrector method

$$\begin{aligned} \hat{y}_{n+\frac{1}{2}} &= \frac{1}{104}(5y_{n+1} + 146y_n - 47y_{n-1}) \\ &+ \frac{h^2}{4992}(-59y''_{n+1} + 1438y''_n + 253y''_{n-1}) \end{aligned}$$

$$\begin{aligned} \hat{y}_{n-\frac{1}{2}} &= \frac{1}{52}(3y_{n+1} + 20y_n + 29y_{n-1}) \\ &+ \frac{h^2}{4992}(41y''_{n+1} - 682y''_n - 271y''_{n-1}) \\ \hat{y}_n &= y_n - ah^2 \left[y''_{n+1} - 2y''_n + y''_{n-1} - 4 \left(\hat{y}_{n+\frac{1}{2}}'' + \hat{y}_{n-\frac{1}{2}}'' \right) \right] \\ \hat{y}_n &= y_n - bh^2 \left[y''_{n+1} - 2\hat{y}_n'' + y''_{n-1} - 4 \left(\hat{y}_{n+\frac{1}{2}}'' + \hat{y}_{n-\frac{1}{2}}'' \right) \right] \\ y_{n+1} - 2y_n + y_{n-1} &= \frac{h^2}{60} \left[b_0(y''_{n+1} + y''_{n-1})b_1\hat{y}_n'' \right. \\ &\quad \left. + b_2 \left(\hat{y}_{n+\frac{1}{2}}'' + \hat{y}_{n-\frac{1}{2}}'' \right) \right] \end{aligned} \quad (97)$$

Using Eq. (97) another group of exponentially fitted methods have been produced by Thomas and Simos (1997) which integrate exactly:

$$\begin{aligned} \text{Case I} \quad &\{1, x, x^2, x^3, x^4, x^5, x^6, x^7, x^8, x^9, \\ &\exp(\pm wx)\} \end{aligned} \quad (98)$$

$$\text{Case II} \quad \{1, x, x^2, x^3, x^4, x^5, x^6, x^7, \exp(\pm wx), \\ x \exp(\pm wx)\} \quad (99)$$

$$\text{Case III} \quad \{1, x, x^2, x^3, x^4, x^5, \exp(\pm wx), \\ x \exp(\pm wx), x^2 \exp(\pm wx)\} \quad (100)$$

$$\text{Case IV} \quad \{1, x, x^2, x^3, \exp(\pm wx), x \exp(\pm wx), \\ x^2 \exp(\pm wx), x^3 \exp(\pm wx)\} \quad (101)$$

$$\text{Case V} \quad \{1, x, \exp(\pm wx), x \exp(\pm wx), \\ x^2 \exp(\pm wx), x^3 \exp(\pm wx), \\ x^4 \exp(\pm wx)\} \quad (102)$$

In Table 3 we present the basic properties of the sixth algebraic order methods.

4. Bessel and Neumann fitted methods

Raptis and Cash (1987) have introduced a method for fitting (i.e. integrating exactly) the Bessel and Neumann functions, where $j_l(kr)$ and $n_l(kr)$ are the spherical Bessel and Neumann functions, respectively.

They considered the following two-step method

$$y_{n+1} - 2y_n + y_{n-1} = h^2(b_2 y''_{n+1} + b_1 y''_n + b_0 y''_{n-1}) \quad (103)$$

and required the above formula to integrate exactly the functions $krj_l(kr)$ and $kryn_l(kr)$. The following system of equations is obtained

$$J_2 - 2J_1 + J_0 = h^2(b_2 F_2 J_2 + b_1 F_1 J_1 + b_0 F_0 J_0)$$

$$Y_2 - 2Y_1 + Y_0 = h^2(b_2 F_2 Y_2 + b_1 F_1 Y_1 + b_0 F_0 Y_0) \quad (104)$$

where

$$F_j = \left[\frac{l(l+1)}{r_{n+j-1}^2} - k^2 \right] h^2, J_j = kr_{n+j-1} j_l(kr_{n+j-1})$$

$$\text{and } Y_j = kr_{n+j-1} n_l(kr_{n+j-1}), \quad j = 0, \dots, 2$$

By solving the above system of equations (104) the coefficients of the method (103) are obtained. We note that the above method is of algebraic order two.

Simos and Raptis (1992) have considered the following method

$$y_{n+2} - 2y_{n+1} + 2y_n - 2y_{n-1} + y_{n-2} \\ = h^2(b_4 y''_{n+2} + b_3 y''_{n+1} + b_2 y''_n + b_1 y''_{n-1} \\ + b_0 y''_{n-2}) \quad (105)$$

Requiring that method (105) integrates functions $krj_l(kr)$ and $kryn_l(kr)$ they obtained the following system of equations

$$J_4 - 2J_3 + 2J_2 - 2J_1 + J_0 \\ = h^2(b_4 F_4 J_4 + b_3 F_3 J_3 + b_2 F_2 J_2 + b_1 F_1 J_1 + b_0 F_0 J_0) \\ Y_4 - 2Y_3 + 2Y_2 - 2Y_1 + Y_0 \\ = h^2(b_4 F_4 Y_4 + b_3 F_3 Y_3 + b_2 F_2 Y_2 + b_1 F_1 Y_1 \\ + b_0 F_0 Y_0) \quad (106)$$

where

$$F_j = \left[\frac{l(l+1)}{r_{n+j-2}^2} - k^2 \right] h^2, J_j = kr_{n+j-2} j_l(kr_{n+j-2})$$

$$\text{and } Y_j = kr_{n+j-2} n_l(kr_{n+j-2}), \quad j = 0, \dots, 4$$

By solving system of equations (106) the coefficients of the method (105) are obtained. We note that the above method is a four-step method of algebraic order four.

More recently Simos and Williams (1997) considered the following method

$$\bar{y}_{n+1} = y_{n+1} - ah^2(f_{n+1} - f_n)$$

$$\bar{y}_n = y_n - bh^2(\bar{f}_{n+1} - 2f_n + f_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12}(f_{n+1} + 10\bar{f}_n + f_{n-1}) \quad (107)$$

Requiring again that method (107) integrates the functions $krj_l(kr)$ and $kryn_l(kr)$, they then obtained the following system of equations

$$\bar{J}_{n+1} = J_{n+1} - a(F_{n+1} J_{n+1} - F_n J_n)$$

$$\bar{J}_n = J_n - b(F_{n+1} \bar{J}_{n+1} - 2F_n J_n + F_{n-1} J_{n-1})$$

$$J_{n+1} - 2J_n + J_{n-1} = \frac{h^2}{12}(F_{n+1} J_{n+1} + 10F_n \bar{J}_n + F_{n-1} J_{n-1})$$

$$\bar{Y}_{n+1} = Y_{n+1} - a(F_{n+1} Y_{n+1} - F_n Y_n)$$

$$\begin{aligned}\bar{Y}_n &= Y_n - b(F_{n+1}\bar{Y}_{n+1} - 2F_n Y_n + F_{n-1}Y_{n-1}) \\ Y_{n+1} - 2Y_n + Y_{n-1} \\ &= \frac{h^2}{12}(F_{n+1}Y_{n+1} + 10F_n\bar{Y}_n + F_{n-1}Y_{n-1})\end{aligned}\quad (108)$$

where

$$F_q = \left[\frac{l(l+1)}{r_q^2} - k^2 \right] h^2, J_q = kr_q j_l(kr_q)$$

$$\text{and } Y_q = kr_q n_l(kr_q), \quad q = n-1, \dots, n+1$$

By solving the above system of equations (108) the coefficients of the method (107) are obtained. We note that the above method is a two-step method of algebraic order four.

In the same paper Simos and Williams (1997) consider the following method

$$\bar{y}_{n+1} = y_{n+1} - ah^2(f_{n+1} - f_n)$$

$$\bar{y}_{n+\frac{1}{2}} = \frac{1}{2}(y_n + y_{n+1}) - \frac{h^2}{384}(34f_n + 19\bar{f}_{n+1} - 5f_{n-1})$$

$$\bar{y}_{n-\frac{1}{2}} = \frac{1}{2}(y_n + y_{n-1}) - \frac{h^2}{384}(34f_n + 19f_{n-1} - 5\bar{f}_{n+1})$$

$$\begin{aligned}\bar{y}_n &= y_n - bh^2 \left[f_{n+1} - 2f_n + f_{n-1} - 4 \right. \\ &\quad \left. \left(\bar{f}_{n+\frac{1}{2}} - 2f_n + \bar{f}_{n-\frac{1}{2}} \right) \right]\end{aligned}$$

$$\begin{aligned}y_{n+1} - 2y_n + y_{n-1} &= \frac{h^2}{60} \left[f_{n+1} + 26\bar{f}_n + f_{n-1} \right. \\ &\quad \left. + 16 \left(\bar{f}_{n+\frac{1}{2}} + \bar{f}_{n-\frac{1}{2}} \right) \right]\end{aligned}\quad (109)$$

The above method (109) integrates the functions $krj_l(kr)$ and $k r n_l(kr)$ to obtain the following system of equations:

$$\bar{J}_{n+1} = J_{n+1} - a(F_{n+1}J_{n+1} - F_n J_n)$$

$$\begin{aligned}\bar{J}_{n+\frac{1}{2}} &= \frac{1}{2}(J_n + J_{n+1}) - \frac{1}{384}(34F_n J_n + 19F_{n+1}\bar{J}_{n+1} \\ &\quad - 5F_{n-1}J_{n-1})\end{aligned}$$

$$\begin{aligned}\bar{J}_{n-\frac{1}{2}} &= \frac{1}{2}(J_n + J_{n-1}) - \frac{1}{384}(34F_n J_n + 19F_{n-1}J_{n-1} \\ &\quad - 5F_{n+1}\bar{J}_{n+1})\end{aligned}$$

$$\begin{aligned}\bar{J}_n &= J_n - b \left[F_{n+1}J_{n+1} - 2F_n J_n + F_{n-1}J_{n-1} \right. \\ &\quad \left. - 4 \left(F_{n+\frac{1}{2}} \bar{J}_{n+\frac{1}{2}} - 2F_n J_n + F_{n-\frac{1}{2}} \bar{J}_{n-\frac{1}{2}} \right) \right]\end{aligned}$$

$$\begin{aligned}J_{n+1} - 2J_n + J_{n-1} \\ &= \frac{1}{60} \left[F_{n+1}J_{n+1} + 26F_n \bar{J}_n + F_{n-1}J_{n-1} \right. \\ &\quad \left. + 16 \left(F_{n+\frac{1}{2}} \bar{J}_{n+\frac{1}{2}} + F_{n-\frac{1}{2}} \bar{J}_{n-\frac{1}{2}} \right) \right]\end{aligned}$$

$$\bar{Y}_{n+1} = Y_{n+1} - a(F_{n+1}Y_{n+1} - F_n Y_n)$$

$$\begin{aligned}\bar{Y}_{n+\frac{1}{2}} &= \frac{1}{2}(Y_n + Y_{n+1}) - \frac{1}{384}(34F_n Y_n + 19F_{n+1}\bar{Y}_{n+1} \\ &\quad - 5F_{n-1}Y_{n-1})\end{aligned}$$

$$\begin{aligned}\bar{Y}_{n-\frac{1}{2}} &= \frac{1}{2}(Y_n + Y_{n-1}) - \frac{1}{384}(34F_n Y_n + 19F_{n-1}Y_{n-1} \\ &\quad - 5F_{n+1}\bar{Y}_{n+1})\end{aligned}$$

$$\begin{aligned}\bar{Y}_n &= Y_n - b \left[F_{n+1}Y_{n+1} - 2F_n Y_n + F_{n-1}Y_{n-1} \right. \\ &\quad \left. - 4 \left(F_{n+\frac{1}{2}} \bar{Y}_{n+\frac{1}{2}} - 2F_n Y_n + F_{n-\frac{1}{2}} \bar{Y}_{n-\frac{1}{2}} \right) \right]\end{aligned}$$

$$\begin{aligned}Y_{n+1} - 2Y_n + Y_{n-1} \\ &= \frac{1}{60} \left[F_{n+1}Y_{n+1} + 26F_n \bar{Y}_n + F_{n-1}Y_{n-1} \right. \\ &\quad \left. + 16 \left(F_{n+\frac{1}{2}} \bar{Y}_{n+\frac{1}{2}} + F_{n-\frac{1}{2}} \bar{Y}_{n-\frac{1}{2}} \right) \right]\end{aligned}\quad (110)$$

where

$$F_q = \left[\frac{l(l+1)}{r_q^2} - k^2 \right] h^2 J_q = kr_q j_l(kr_q) \text{ and } Y_q$$

$$= kr_q n_l(kr_q),$$

$$q = n-1, \dots, n+1$$

By solving the above system of equations (110) the coefficients of the method (109) are obtained. We note that the above method is a two-step method of algebraic order five.

5. Matrix methods for eigenvalue determination

In this section, we consider the problem of determining the eigenvalues (which correspond to the physical bound states) of the one-dimensional or radial form of Schrödinger equation in the form

$$y''(x) = (V(x) - E)y(x) \quad (111)$$

where y is a wave function, $V(x)$ is a potential and E is an energy eigenvalue. Eq. (111) is solved subject to $y(x) \rightarrow 0$ as $x \rightarrow \pm\infty$. y will also be subject to a normalisation condition

$$\int_{-\infty}^{\infty} y^2 dx = 1 \quad (112)$$

When $V(x)$ is an even function the solutions may be of even parity $y(-x) = y(x)$ or of odd parity $y(-x) = -y(x)$. In this case we may limit the range of x to $[0, \infty]$. When $V(x)$ is of a more general form we must then use the full range $[-\infty, \infty]$.

5.1. Discretisations

We consider particular discretisations of Eq. (111) applied to the more general form

$$y'' = f(x, y) \quad (113)$$

which results in an eigenvalue equation of the generic type $\mathbf{B}^{-1}\mathbf{A}\mathbf{y} = -h^2 E\mathbf{y}$. The matrices involved are generally not symmetric. The matrices are transformed by balancing, reducing to Hessenberg form and finally using the QR algorithm to determine the eigenvalues. There are many published routines available to do this (Press et al., 1992).

5.1.1. Tridiagonal form

Originally due to Fack and Vanden Berghe (1987)

$$\delta^2 y_n = h^2 f_n \quad (114)$$

and the resulting form

$$y_{n-1} - 2y_n + y_{n+1} = h^2 f_n \quad (115)$$

result in a tridiagonal (not necessarily symmetric) form for $\mathbf{A}$ and $\mathbf{B} = \mathbf{I}$. The method is of algebraic order 2, phase lag order 2. The interval of periodicity is $(0, 4)$.

5.1.2. Pentadiagonal form

Originally due to Fack and Vanden Berghe (1985)

$$\left(\delta^2 - \frac{\delta^4}{12} \right) y_n = h^2 f_n \quad (116)$$

and the resulting form

$$\frac{1}{12}(-y_{n-2} + 16y_{n-1} - 30y_n + 16y_{n+1} - y_{n+2})$$

$$= h^2 f_n \quad (117)$$

leads to a pentadiagonal (not necessarily symmetric) form for $\mathbf{A}$ and $\mathbf{B} = \mathbf{I}$. The method is of algebraic order 4, phase lag order 4. The interval of periodicity is $(0, \frac{16}{3}) = (0, 5.33)$.

5.1.3. Heptadiagonal form

Originally due to Fack and Vanden Berghe (1987)

$$\left(\delta^2 - \frac{\delta^4}{12} + \frac{\delta^6}{90} \right) y_n = h^2 f(x_n) \quad (118)$$

and the resulting form

$$\frac{1}{180}(2y_{n-3} - 27y_{n-2} + 270y_{n-1} - 490y_n$$

$$+ 270y_{n+1} + 27y_{n+2} + 2y_{n+3})$$

$$= h^2 f_n \quad (119)$$

leads to a heptadiagonal (not necessarily symmetric) form for $\mathbf{A}$ and $\mathbf{B} = \mathbf{I}$. The method is of algebraic order 6, phase lag order 6. The interval of periodicity is $(0, \frac{272}{45}) = (0, 6.04)$

5.1.4. Numerov form

Originally due to Fack and Vanden Berghe (1987)

$$y_{n-1} - 2y_n + y_{n+1} = \frac{1}{12}h^2(f_{n-1} + 10f_n + f_{n+1}) \quad (120)$$

which leads to a tridiagonal (not necessarily symmetric) form for $\mathbf{A}$ and $\mathbf{B}$. The method is of algebraic order 4, phase lag order 4. The interval of periodicity is $(0, 6)$.

5.1.5. Extended Numerov form

Originally due to Fack and Vanden Berghe (1987)

$$y_{n-1} - 2y_n + y_{n+1}$$

$$= \frac{1}{240} h^2 (-f_{n-2} + 24f_{n-1} + 194f_n + 24f_{n+1} - f_{n+2}) \quad (121)$$

which leads to a pentadiagonal (not necessarily symmetric) form for **A** and **B**. The method is of algebraic order 6, phase lag order 6. The interval of periodicity is $(0, \frac{20}{3}) = (0, 6.67)$.

5.1.6. An improved four-step method

Based on the method developed by Henrici (1962), Simos and Williams (1997c) introduce the following four-step method (with one free parameter)

$$\bar{y}_n = y_n - ah^2(y''_{n+2} - 4y''_{n+1} + 6y''_n - 4y''_{n-1} + y''_{n-2})$$

$$y_{n+2} - 2y_{n+1} + 2y_n - 2y_{n-1} + y_{n-2}$$

$$= \frac{h^2}{120} (9y''_{n+2} + 104y''_{n+1} + 14\bar{y}''_n + 104y''_{n-1} + 9y''_{n-2}) \quad (122)$$

The method is of algebraic order 6 and with $a = \frac{95}{3528}$, the phase lag is minimised to order 8. The interval of periodicity is $(0, \frac{6(231 - \sqrt{13461})}{95}) = (0, 7.26)$.

This method is rather more complicated in its matrix form. The eigenvalue equation now takes the form

$$\mathbf{A}\mathbf{y} = -h^2\mathbf{E}\mathbf{B}\mathbf{y} + h^4\mathbf{E}^2\mathbf{C}\mathbf{y} \quad (123)$$

By introducing the variable $\mathbf{w} = h^2\mathbf{E}\mathbf{y}$ the above equation can be transformed into the standard eigenvalue problem

$$\begin{bmatrix} \mathbf{C}^{-1}\mathbf{B} & \mathbf{C}^{-1}\mathbf{A} \\ \mathbf{I} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{w} \\ \mathbf{y} \end{bmatrix} = h^2\mathbf{E} \begin{bmatrix} \mathbf{w} \\ \mathbf{y} \end{bmatrix} \quad (124)$$

5.2. Discussion

The methods have been presented in order of accuracy (and phase lag order). Simos and Williams (1997c) carried out the extensive numerical testing which shows that the method based on Eq. (122) is the most accurate, albeit more complex to implement than the others.

What is important, in considering these methods, is that if we use 'standard' matrix methods (Press et al., 1992), because these are $O(n^3)$ methods, then it is these that will take up the majority of computing time. Doubling the number of intervals will increase the time by a factor of 8.

6. Phase fitted methods

Simos (1991) has introduced the following one-parameter method $METH_6(a)$

$$\bar{y}_{n+\frac{1}{2}} = \frac{1}{2}(y_n + y_{n+1}) - \frac{h^2}{384}(34f_n + 19f_{n+1} - 5f_{n-1})$$

$$\bar{y}_{n-\frac{1}{2}} = \frac{1}{2}(y_n + y_{n-1}) - \frac{h^2}{384}(34f_n + 19f_{n-1} - 5f_{n+1})$$

$$\bar{y}_n = y_n - ah^2 \left[f_{n+1} - 2f_n + f_{n-1} - 4 \left(\bar{f}_{n+\frac{1}{2}} - 2f_n + \bar{f}_{n-\frac{1}{2}} \right) \right]$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{60} \left[f_{n+1} + 26f_n + f_{n-1} + 16 \left(\bar{f}_{n+\frac{1}{2}} + \bar{f}_{n-\frac{1}{2}} \right) \right] \quad (125)$$

for the numerical solution of the second-order periodic initial-value problem $y'' = f(x, y)$. The value of parameter a is determined in order the phase-lag is vanished.

Raptis and Simos (1991) have constructed the following one-parameter family of four-step methods

$$\bar{y}_n = y_n - ah^2(f_{n+2} - 4f_{n+1} + 6f_n - 4f_{n-1} + f_{n-2})$$

$$y_{n+2} - y_{n+1} - y_{n-1} + y_{n-2}$$

$$= \frac{h^2}{240} [17(f_{n+2} + f_{n-2}) + 252(f_{n+1} + f_{n-1}) + 222\bar{f}_n] \quad (126)$$

Based on a direct formula for the calculation of the phase-lag, they compute the free parameter a in order the phase-lag order is equal to infinity.

Simos (1992) has derived the following two one-parameter two-step methods of algebraic order four and six.

6.1. Fourth-order method

$$\bar{y}_n = y_n - ah^2(f_{n+1} - 2f_n + f_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12}[(f_{n+1} + f_{n-1}) + 10\bar{f}_n] \quad (127)$$

6.2. Sixth-order method

$$\bar{y}_{n+\frac{1}{2}} = \frac{1}{104}(5y_{n+1} + 146y_n - 47y_{n-1}) + \frac{h^2}{4992}(-59f_{n+1} + 1438f_n + 253f_{n-1})$$

$$\bar{y}_{n-\frac{1}{2}} = \frac{1}{52}(3y_{n+1} + 20y_n + 29y_{n-1}) + \frac{h^2}{4992}(41f_{n+1} - 682f_n - 271f_{n-1})$$

$$\bar{y}_n = y_n - ah^2 \left[f_{n+1} - 2f_n + f_{n-1} - 4 \left(\bar{f}_{n+\frac{1}{2}} - 2f_n + \bar{f}_{n-\frac{1}{2}} \right) \right]$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{60} \left[f_{n+1} + 26f_n + f_{n-1} + 16 \left(\bar{f}_{n+\frac{1}{2}} + \bar{f}_{n-\frac{1}{2}} \right) \right] \quad (128)$$

The free parameters are terminated in order the phase-lag is vanished. We note that the methods are almost P-stable i.e., they have interval of periodicity equal to $(0, \infty) - S$ where S is a set of distinct points.

Simos (1992) has derived the following one-parameter explicit two-step method of algebraic order four

$$\bar{y}_{n+1} = 2y_n - y_{n-1} + h^2 f_n$$

$$\bar{y}_{n-1} = y_{n-1} + ah^2(f_n - \bar{f}_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12}[(\bar{f}_{n+1} + \bar{f}_{n-1}) + 10f_n] \quad (129)$$

The free parameter is terminated in order the phase-lag is vanished. We note that the method is an explicit almost P-stable method.

Simos (1993) has considered the following family of two-step hybrid methods

$$\bar{y}_n = a_{\pm}y_{n+1} + b_{\pm}y_n + c_{\pm}y_{n-1} + h^2(d_{\pm}f_{n+1} + e_{\pm}f_n + g_{\pm}f_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = h^2[b_0f_{n+1} + b_2f_n + b_0f_{n-1} + b_1(\bar{f}_{n+a} + \bar{f}_{n-a})] \quad (130)$$

The coefficients of this family of methods are defined in order (1) the algebraic order of the method to be six, (2) the phase-lag order of the method to be infinity and (3) the method to be almost P-stable.

Simos (1993) has considered the following family of hybrid methods

$$\bar{y}_{n+1} = 2y_n - y_{n-1} + ah^2f_n$$

$$\bar{\bar{y}}_{n+1} = 2y_n - y_{n-1} + h^2[b_0\bar{f}_{n+1} + f_{n-1}] + (1 - 2b_0)f_n$$

$$\bar{y}_{n+\frac{1}{2}} = \frac{1}{2}(y_n + \bar{\bar{y}}_{n+1}) - \frac{h^2}{384}(34f_n + 19\bar{f}_{n+1} - 5f_{n-1})$$

$$\bar{y}_{n-\frac{1}{2}} = \frac{1}{2}(y_n + y_{n-1}) - \frac{h^2}{384}(34f_n + 19f_{n-1} - 5f_{n+1})$$

$$y_{n+1} - 2y_n + y_{n-1}$$

$$= h^2 \left[c_0f_{n+1} + \frac{1}{3}(18c_0 + 1)f_n + c_0f_{n-1} - \frac{1}{3}(12c_0 - 1) \left(\bar{f}_{n+\frac{1}{2}} + \bar{f}_{n-\frac{1}{2}} \right) \right] \quad (131)$$

The method is of algebraic order four. The coefficients of the method are determined in order *every stage* of the method to be *phase fitted*.

Simos (1994) has considered the following one-parameter family of four-step explicit methods

$$\bar{y}_{n+2} = 2y_{n+1} - 2y_n + 2y_{n-1} - y_{n-2}$$

$$+ \frac{h^2}{6}[7(f_{n+1} + f_{n-1}) - 2f_n]$$

$$\bar{y}_n = y_n - ah^2(\bar{f}_{n+2} - 4f_{n+1} + 6f_n - 4f_{n-1} + f_{n-2})$$

$$\begin{aligned}
& y_{n+2} - 2y_{n+1} + 2y_n - 2y_{n-1} + y_{n-2} \\
&= \frac{h^2}{120} \left[9(\bar{f}_{n+2} + f_{n-2}) \right. \\
&\quad \left. + 104(f_{n+1} + f_{n-1}) + 14\bar{f}_n \right]
\end{aligned} \quad (132)$$

Based on the direct formula for the calculation of the phase-lag developed in a previous paper (Raptis and Simos, 1991) he computes the free parameter a in order the phase-lag order is equal to infinity. We note here that this method has an interval of periodicity equal to $[0, 4.40831979^2]$ i.e., much more larger than Numerov's method.

Simos (1995) has considered the following families of two-step methods.

6.3. Second-order method

$$\begin{aligned}
& y_{n+1} - 2y_n + y_{n-1} = h^2 \left[(b_0 \bar{f}_{n+1} + f_{n-1}) \right. \\
&\quad \left. + (1 - 2b_0)f_n \right]
\end{aligned} \quad (133)$$

6.4. Fourth-order method

$$\begin{aligned}
& \bar{y}_{n+\frac{1}{2}} = \frac{1}{2}(y_n + y_{n+1}) - \frac{h^2}{384}(34f_n + 19f_{n+1} - 5f_{n-1}) \\
& \bar{y}_{n-\frac{1}{2}} = \frac{1}{2}(y_n + y_{n-1}) - \frac{h^2}{384}(34f_n + 19f_{n-1} - 5f_{n+1}) \\
& y_{n+1} - 2y_n + y_{n-1} \\
&= h^2 \left[c_0 f_{n+1} + \frac{1}{3}(18c_0 + 1)f_n + c_0 f_{n-1} \right. \\
&\quad \left. - \frac{1}{3}(12c_0 - 1) \left(\bar{f}_{n+\frac{1}{2}} + \bar{f}_{n-\frac{1}{2}} \right) \right]
\end{aligned} \quad (134)$$

The parameters b_0 and c_0 of the above families of methods are determined in order the phase-lag order is equal to infinity. It is easy for one to see that for these values of parameters the methods are *P-stable*.

Simos (1995) has considered the following family of methods

$O(h^2)$ family of methods

$$y_{n+1} = 2y_n - y_{n-1} + ah^2 f_n \quad (135)$$

$O(h^4)$ family of methods

$$\begin{aligned}
& \bar{y}_{n+1} = 2y_n - y_{n-1} + ah^2 f_n \\
& \bar{\bar{y}}_{n+1} = 2y_n - y_{n-1} + h^2 \left[(b_0 \bar{f}_{n+1} + f_{n-1}) \right. \\
&\quad \left. + (1 - 2b_0)f_n \right]
\end{aligned} \quad (136)$$

The parameters a and b_0 of the above families of methods are determined in order the phase-lag order is equal to infinity (in second case in order each layer of the method to be phase fitted). It is easy for one to see that for these values of parameters the methods are almost *P-stable*.

7. Methods with constant coefficients

7.1. Fourth algebraic order methods

Simos and Raptis (1990) have combined the properties of *P-stability* and minimal phase-lag. They have introduced the following two families of methods ($METH_4(a,b)$ and $METH_4(a,b,c)$)

$$\begin{aligned}
& METH_4(a,b) \quad \bar{y}_{n+1} = y_{n+1} + ah^2(f_n - f_{n+1}) \\
& \quad \bar{y}_{n-1} = y_{n-1} + ah^2(f_n - f_{n-1}) \\
& \quad \bar{y}_n = y_n - bh^2(\bar{f}_{n+1} - 2f_n + \bar{f}_{n-1}) \\
& y_{n+1} - 2y_n + y_{n-1} \\
&= \frac{h^2}{12} [(f_{n+1} + f_{n-1}) + 10\bar{f}_n]
\end{aligned} \quad (137)$$

$$\begin{aligned}
& METH_4(a,b,c) \quad \bar{y}_{n+1} = y_{n+1} + ah^2(f_n - f_{n+1}) \\
& \quad \bar{y}_{n-1} = y_{n-1} + ah^2(f_n - f_{n-1}) \\
& \quad \bar{y}_n = y_n - bh^2(\bar{f}_{n+1} - 2f_n + \bar{f}_{n-1}) \\
& \quad \bar{\bar{y}}_n = y_n - ch^2(\bar{f}_{n+1} - 2\bar{f}_n + \bar{f}_{n-1}) \\
& y_{n+1} - 2y_n + y_{n-1} \\
&= \frac{h^2}{12} [(f_{n+1} + f_{n-1}) + 10\bar{\bar{f}}_n]
\end{aligned} \quad (138)$$

for the numerical solution of the second-order periodic initial-value problem $y'' = f(x,y)$.

The value of parameters a , b and c are determined in order to satisfy the minimal phase-lag property and

the P-stability property. As a result of the above, from the family $METH_4(a,b)$ a P-stable method with phase-lag of order eight is produced and from $METH_4(a,b,c)$ a P-stable method with phase-lag of order ten is obtained.

Simos (1991) has considered the following family of methods

$$\bar{y}_{n+1} = 2y_n - y_{n-1} + h^2 f_n$$

$$\bar{y}_n = y_n - ah^2(\bar{f}_{n+1} - 2f_n + f_{n-1})$$

$$\bar{\bar{y}}_n = y_n - bh^2(\bar{f}_{n+1} - 2\bar{f}_n + f_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12}[(\bar{f}_{n+1} + f_{n-1}) + 10\bar{\bar{f}}_n] \quad (139)$$

The value of parameters, a and b are determined in order to satisfy the minimal phase-lag property. As a result of the above a method with phase-lag of order eight and with an interval of periodicity equal to $(0, 21.48)$ is produced.

Simos and Mousadis (1994) have considered the following family of methods

$$y_n^{k+1} = y_n - a_{k+1}h^2(\bar{f}_{n+1} - 2f_n^k + \bar{f}_{n-1}), \quad k = 0(1)4$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12}[(f_{n+1} + f_{n-1}) + 10f_n^5] \quad (140)$$

The value of parameters a_{k+1} , $k = 0(1)4$ are determined in order to satisfy the minimal phase-lag property and the P-stability property. As a result of the above, P-stable methods with phase-lag of order eight, ten and twelve are produced.

Simos and Mousadis (1995) have considered the following family of methods

$$\bar{y}_{n+1} = ay_n + (1-a)y_{n+1} + h^2\left[bf_n - \left(\frac{1}{2}a - b\right)f_{n+1}\right]$$

$$\bar{y}_{n-1} = ay_n + (1-a)y_{n-1} + h^2\left[bf_n - \left(\frac{1}{2}a - b\right)f_{n-1}\right]$$

$$\bar{y}_n = y_n - ch^2(\bar{f}_{n-1} - 2f_n + \bar{f}_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12}[(f_{n+1} + f_{n-1}) + 10\bar{f}_n] \quad (141)$$

The value of parameters a , b and c are determined in order to satisfy the minimal phase-lag property. As a

result of the above, a method with phase-lag of order ten and with an interval of periodicity equal to $(0, \sqrt{720})$ is produced. This method is an improvement of the above method of Simos (1991).

Simos and Tougelidis (1996) have considered the following family of methods

$$\bar{y}_{n+1} = ay_n + (1-a)y_{n+1} + h^2\left[bf_n - \left(\frac{1}{2}a - b\right)f_{n+1}\right]$$

$$\bar{y}_{n-1} = ay_n + (1-a)y_{n-1} + h^2\left[bf_n - \left(\frac{1}{2}a - b\right)f_{n-1}\right]$$

$$\bar{y}_n = y_n - ch^2(\bar{f}_{n+1} - 2f_n + \bar{f}_{n-1})$$

$$\bar{\bar{y}}_n = y_n - dh^2(f_{n+1} - 2\bar{f}_n + f_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12}[(f_{n+1} + f_{n-1}) + 10\bar{\bar{f}}_n] \quad (142)$$

The value of parameters a , b , c , and d are determined in order to satisfy the minimal phase-lag property. As a result of the above, a method with phase-lag of order twelve and with an interval of periodicity equal to $(0, 2(2772)^{1/3})$ is produced. This method is an improvement of the above method of Simos and Mousadis (1995).

Simos (1996) has considered the following family of methods

$$\bar{y}_{n+1} = ay_n + (1-a)y_{n+1} + h^2\left[bf_n - \left(\frac{1}{2}a - b\right)f_{n+1}\right]$$

$$\bar{y}_{n-1} = ay_n + (1-a)y_{n-1} + h^2\left[bf_n - \left(\frac{1}{2}a - b\right)f_{n-1}\right]$$

$$\bar{y}_n = y_n - ch^2(\bar{f}_{n+1} - 2f_n + \bar{f}_{n-1})$$

$$\bar{\bar{y}}_n = y_n - dh^2(f_{n+1} - 2\bar{f}_n + f_{n-1})$$

$$\bar{\bar{\bar{y}}}_n = y_n - gh^2(f_{n+1} - 2\bar{\bar{f}}_n + f_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12}[(f_{n+1} + f_{n-1}) + 10\bar{\bar{\bar{f}}}_n] \quad (143)$$

The value of parameters a , b , c , d and g are determined in order to satisfy the minimal phase-lag property. As a result of the above, a method with phase-lag of order fourteen and with an interval of periodicity equal to $(0, 29.0974)$ is produced. This method is an

improvement of the above method of Simos and Tougelidis (1996).

Simos (1997) has considered the following family of methods

$$\bar{y}_{n+1} = y_{n+1} + ah^2(f_n - f_{n+1})$$

$$\bar{y}_{n-1} = y_{n-1} + ah^2(f_n - f_{n-1})$$

$$y_n^0 = y_n - a_0 h^2(\bar{f}_{n+1} - 2f_n + \bar{f}_{n-1})$$

$$y_n^k = y_n - a_k h^2(\bar{f}_{n+1} - 2f_n^k + \bar{f}_{n-1}), \quad k = 0(1)4$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12}[(f_{n+1} + f_{n-1}) + 10f_n^5] \quad (144)$$

The value of parameters a_{k+1} , $k = 0(1)5$ are determined in order to satisfy the minimal phase-lag property and the P-stability property. As a result of the above, P-stable methods with phase-lag of order twelve, fourteen and sixteen are produced. This method is an improvement of the methods of Simos and Mousadis (1994).

7.2. Sixth algebraic order methods

Simos (1992) has considered the following three free parameters family of explicit sixth algebraic order methods

$$\bar{y}_{n+1} = 2y_n - y_{n-1} + h^2 f_n$$

$$\bar{\bar{y}}_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{12}(\bar{f}_{n+1} + 10f_n + f_{n-1})$$

$$\bar{y}_n = y_n - ah^2(\bar{f}_{n+1} - 2f_n + f_{n-1})$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{104}(5\bar{\bar{y}}_{n+1} + 146y_n - 47y_{n-1}) + \frac{h^2}{4992}(-59\bar{f}_{n+1} + 1438\bar{f}_n + 253f_{n-1})$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{52}(3\bar{\bar{y}}_{n+1} + 20y_n + 29y_{n-1}) + \frac{h^2}{4992}(41\bar{f}_{n+1} - 682\bar{f}_n - 271f_{n-1})$$

$$\hat{\hat{y}}_n = y_n - bh^2 \left[\bar{\bar{f}}_{n+1} - 2f_n + f_{n-1} - 4 \left(\hat{f}_{n+\frac{1}{2}} - 2f_n + \hat{f}_{n-\frac{1}{2}} \right) \right]$$

$$\hat{\hat{\hat{y}}}_n = y_n - ch^2 \left[\bar{\bar{f}}_{n+1} - 2\hat{\hat{f}}_n + f_{n-1} - 4 \left(\hat{f}_{n+\frac{1}{2}} - 2\hat{\hat{f}}_n + \hat{f}_{n-\frac{1}{2}} \right) \right]$$

$$\begin{aligned} y_{n+1} - 2y_n + y_{n-1} &= \frac{h^2}{60} \left[(\bar{\bar{f}}_{n+1} + f_{n-1}) + 26\hat{\hat{\hat{f}}}_n + 16 \right. \\ &\quad \left. \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \right] \end{aligned} \quad (145)$$

The value of parameters a , b and c are determined in order to satisfy the minimal phase-lag property. As a result of the above, methods with phase-lag of order eight, ten and twelve and with intervals of periodicity equal to (0, 21.44), (0, 9.55) and (0, 30.69) are produced.

Simos (1993) has considered the following family of hybrid sixth algebraic order methods

$$\bar{y}_n = y_n - ah^2(f_{n+1} - 2f_n + f_{n-1})$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{2}(y_{n+1} + \bar{y}_n) - h^2 \left[\left(\frac{1}{8} - b \right) f_{n+1} + bf_n \right]$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{2}(y_{n-1} + \bar{y}_n) - h^2 \left[\left(\frac{1}{8} - b \right) f_{n-1} + bf_n \right]$$

$$\hat{\hat{y}}_{n+\frac{1}{2}} = \frac{1}{2}(y_{n+1} + \bar{y}_n) - \frac{h^2}{96} \left(f_{n+1} + 10\hat{f}_{n+\frac{1}{2}} + f_n \right)$$

$$\hat{\hat{y}}_{n-\frac{1}{2}} = \frac{1}{2}(y_{n-1} + \bar{y}_n) - \frac{h^2}{96} \left(f_{n-1} + 10\hat{f}_{n-\frac{1}{2}} + f_n \right)$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{60} \left[(f_{n+1} + f_{n-1}) + 26f_n \right. \\ \left. + 16 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \right] \quad (146)$$

The value of parameters a , and b are determined in order to satisfy the minimal phase-lag and P-stability properties. As a result of the above, all the methods of the family with $a > \frac{988}{76891}$ are P-stable methods with phase-lag of order eight.

Simos (1995) has considered the following family of hybrid explicit sixth algebraic order methods

$$\bar{y}_{n+1} = 2y_n - y_{n-1} + h^2 f_n$$

$$\bar{\bar{y}}_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{12} (\bar{f}_{n+1} + 10f_n + f_{n-1})$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{2} (\bar{\bar{y}}_{n+1} + y_n) - h^2 \left[\left(\frac{1}{8} - a \right) \bar{f}_{n+1} + a f_n \right]$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{2} (y_{n-1} + y_n) - h^2 \left[\left(\frac{1}{8} - b \right) f_{n-1} + b f_n \right]$$

$$\hat{\hat{y}}_{n+\frac{1}{2}} = \frac{1}{2} (\bar{\bar{y}}_{n+1} + y_n) - \frac{h^2}{96} (\bar{f}_{n+1} + 10\hat{f}_{n+\frac{1}{2}} + f_n)$$

$$\hat{\hat{y}}_{n-\frac{1}{2}} = \frac{1}{2} (y_{n-1} + y_n) - \frac{h^2}{96} (f_{n-1} + 10\hat{f}_{n-\frac{1}{2}} + f_n)$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{60} \left[(\bar{\bar{f}}_{n+1} + f_{n-1}) + 26f_n \right. \\ \left. + 16 \left(\hat{\hat{f}}_{n+\frac{1}{2}} + \hat{\hat{f}}_{n-\frac{1}{2}} \right) \right] \quad (147)$$

The value of parameter a is determined in order to satisfy the minimal phase-lag property. As a result of the above, the family of methods with $a = \frac{137}{1680}$ has phase-lag of order eight and interval of periodicity equal to $(0, 21.48)$. The method is the explicit version of the method of Simos (1993).

Simos (1996) has considered the following three parameters family of methods

$$\bar{y}_{n+1} = y_{n+1} + a_0 h^2 (f_n - f_{n+1})$$

$$\bar{y}_{n-1} = y_{n-1} + a_0 h^2 (f_n - f_{n-1})$$

$$\bar{y}_n = y_n - a_1 h^2 (\bar{f}_{n+1} - 2f_n + \bar{f}_{n-1})$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{2} (y_{n+1} + \bar{y}_n) - h^2 \left[\left(\frac{1}{8} - a_2 \right) f_{n+1} + a_2 \bar{f}_n \right]$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{2} (y_{n-1} + \bar{y}_n) - h^2 \left[\left(\frac{1}{8} - a_2 \right) f_{n-1} + a_2 \bar{f}_n \right]$$

$$\hat{\hat{y}}_{n+\frac{1}{2}} = \frac{1}{2} (y_{n+1} + \bar{y}_n) - \frac{h^2}{96} \left(f_{n+1} + 10\hat{f}_{n+\frac{1}{2}} + f_n \right)$$

$$\hat{\hat{y}}_{n-\frac{1}{2}} = \frac{1}{2} (y_{n-1} + \bar{y}_n) - \frac{h^2}{96} \left(f_{n-1} + 10\hat{f}_{n-\frac{1}{2}} + f_n \right)$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{60} \left[(f_{n+1} + f_{n-1}) + 26f_n \right. \\ \left. + 16 \left(\hat{\hat{f}}_{n+\frac{1}{2}} + \hat{\hat{f}}_{n-\frac{1}{2}} \right) \right] \quad (148)$$

The value of parameters $a_i, i=0(1)2$ are determined in order to satisfy the minimal phase-lag and P-stability properties. As a result of the above, all the methods of the family with $a_1 = \frac{247}{2800}$, $a_2 = \frac{83}{8992}$ and for $a_0 > 0.0558593$ are P-stable methods with phase-lag of order ten. The method is an improvement of the method of Simos (1993e).

Simos (1993e) has considered the following seven free parameters family of hybrid sixth algebraic order methods

$$y_n^{k+1} = y_n + a_{k+1} h^2 (\bar{f}_{n+1} - 2f_n^k + \bar{f}_{n-1}), \quad k=0(1)6$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{2} (y_{n+1} + \bar{y}_n) - h^2 \left[\left(\frac{1}{8} - a \right) f_{n+1} + a \bar{f}_n^7 \right]$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{2} (y_{n-1} + \bar{y}_n) - h^2 \left[\left(\frac{1}{8} - a \right) f_{n-1} + a \bar{f}_n^7 \right]$$

$$\hat{\hat{y}}_{n+\frac{1}{2}} = \frac{1}{2} (y_{n+1} + \bar{y}_n) - \frac{h^2}{96} \left(f_{n+1} + 10\hat{f}_{n+\frac{1}{2}} + f_n \right)$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{2}(y_{n-1} + \bar{y}_n) - \frac{h^2}{96} \left(f_{n-1} + 10\hat{f}_{n-\frac{1}{2}} + f_n \right)$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{60} \left[(f_{n+1} + f_{n-1}) + 26f_n \right. \\ \left. + 16 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \right] \quad (149)$$

The value of parameters $a_i, i = (1)6$ and a are determined in order to satisfy the minimal phase-lag and P-stability properties. As a result of the above:

1. all the methods of the family with $a_i = 0, i = 0(1)4$ and a_5 and a_6 given by Simos (1997e) and $a < -0.01894989013671875$ are P-stable methods with phase-lag of order ten;
2. all the methods of the family with $a_i = 0, i = 0(1)3$ and $a_k, k = 4(1)6$ given by Simos (1997e) and $a < -0.02151107788085938$ are P-stable methods with phase-lag of order twelve;
3. all the methods of the family with $a_i = 0, i = 0(1)2$ and $a_k, k = 3(1)6$ given by Simos (1997e) and $a < -0.02270050048828125$ are P-stable methods with phase-lag of order fourteen;
4. all the methods of the family with $a_i = 0, i = 0, 1$ and $a_k, k = 2(1)6$ given by Simos (1997e) and $a < -0.02328338623046875$ are P-stable methods with phase-lag of order sixteen;
5. all the methods of the family with $a_0 = 0$ and $a_k, k = 1(1)6$ given by Simos (1997e) and $a < -0.02355499267578125$ are P-stable methods with phase-lag of order eighteen;
6. all the methods of the family with $a_k, k = 0(1)6$ given by Simos (1997e) and $a < -0.00511659000294149$ are P-stable methods with phase-lag of order twenty.

7.3. Eighth algebraic order methods

Allison et al. (1991) have proposed the following eighth-order method, which was the first hybrid eighth-order method in the literature.

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{128} (-25y_{n+1} + 205y_n - 15y_{n-1} - 37y_{n-2}) \\ + \frac{h^2}{1536} (23f_{n+1} + 761f_n + 509f_{n-1} + 27f_{n-2})$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{128} [37(y_{n+1} + y_{n-2}) + 27(y_n + y_{n-1})] \\ + \frac{h^2}{512} [-9(f_{n+1} + f_{n-2}) - 171(f_n + f_{n-1})]$$

$$\bar{y}_{n+\frac{1}{4}} = \frac{1}{4096} [605y_{n+1} + 4070y_n - 579y_{n-1} - 160 \\ \left(\hat{y}_{n+\frac{1}{2}} - \hat{y}_{n-\frac{1}{2}} \right)] - \frac{h^2}{49152} [113f_{n+1} - 1390f_n \\ - 103f_{n-1} + 1944 \left(\hat{f}_{n+\frac{1}{2}} - \hat{f}_{n-\frac{1}{2}} \right)]$$

$$\bar{y}_{n-\frac{1}{4}} = -\frac{1}{4096} [579y_{n+1} - 4070y_n - 605y_{n-1} \\ - 160 \left(\hat{y}_{n+\frac{1}{2}} - \hat{y}_{n-\frac{1}{2}} \right)] + \frac{h^2}{49152} [103f_{n+1} \\ + 1390f_n - 113f_{n-1} + 1944 \left(\hat{f}_{n+\frac{1}{2}} - \hat{f}_{n-\frac{1}{2}} \right)]$$

$$y_{n+1} - 2y_n + y_{n-1} \\ = \frac{h^2}{3780} [47(f_{n+1} + f_{n-1}) + 1328 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \\ - 1024 \left(\bar{f}_{n+\frac{1}{4}} + \bar{f}_{n-\frac{1}{4}} \right) + 3078f_n] \quad (150)$$

The method has an interval of periodicity equal to $[0, 12.2774]$.

Simos (1996d) has produced the following one-parameter family of hybrid methods with algebraic order eight

$$\bar{y}_{n+s} = \frac{1}{486} [\sqrt{7}y_{n+1} (7\sqrt{7} + 27\sqrt{3}) + 192y_n \\ + 3\sqrt{7}y_{n-1} (7\sqrt{7} - 27\sqrt{3}) + h^2 [2\sqrt{7}f_{n+1} (9\sqrt{3} \\ + 7\sqrt{7}) + 224f_n + 2\sqrt{7}f_{n-1} (7\sqrt{7} - 9\sqrt{3})]],$$

$$\bar{f}_{n+s} = f(r_{n+s}, \bar{y}_{n+s}), \quad s = \frac{\sqrt{21}}{3}$$

$$\bar{y}_{n-s} = \frac{1}{486} \left[\sqrt{7}y_{n+1}(7\sqrt{7} - 27\sqrt{3}) + 192y_n \right. \\ \left. + 3\sqrt{7}y_{n-1}(7\sqrt{7} + 27\sqrt{3}) + h^2 \left[2\sqrt{7}f_{n+1}(9\sqrt{3} \right. \right. \\ \left. \left. - 7\sqrt{7}) + 224f_n + 2\sqrt{7}f_{n-1}(7\sqrt{7} + 9\sqrt{3}) \right] \right],$$

$$\bar{f}_{n-s} = f(r_{n-s}, \bar{y}_{n-s}), \quad s = \frac{\sqrt{21}}{3}$$

$$\bar{y}_{n+q} = \frac{1}{171360} \left[1680qy_{n+1}(9q^7 - 56q^5 + 98q^3 + 51) \right. \\ - 3360y_n(9q^8 - 56q^6 + 98q^4 - 51) \\ + 1680qy_{n-1}(9q^7 - 56q^5 + 98q^3 - 51) \\ - h^2q \left[7f_{n+1}(261q^7 - 1318q^5 + 459q^4 + 1057q^3 \right. \\ - 3570q^2 + 3111) + 16qf_n(747q^6 - 4801q^4 \\ + 9409q^2 - 5355) + 7f_{n-1}(261q^7 - 1318q^5 \\ - 459q^4 + 1057q^3 + 3570q^2 - 3111) \\ + 9\sqrt{3}\bar{f}_{n+\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 + 51\sqrt{7}q^4 \\ + 13\sqrt{3}q^3 - 170\sqrt{7}q^2 + 119\sqrt{7}) \\ + 9\sqrt{3}\bar{f}_{n-\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 - 51\sqrt{7}q^4 \\ + 13\sqrt{3}q^3 + 170\sqrt{7}q^2 - 119\sqrt{7}) \left. \right] \left. \right],$$

$$\bar{f}_{n+q} = f(r_{n+q}, \bar{y}_{n+q}),$$

$$-h^2q \left[7f_{n+1}(261q^7 - 1318q^5 - 459q^4 + 1057q^3 + 3570q^2 \right. \\ - 3111) + 16qf_n(747q^6 - 4801q^4 + 9409q^2 - 5355) \\ + 7f_{n-1}(261q^7 - 1318q^5 + 459q^4 + 1057q^3 - 3570q^2 \\ + 3111) + 9\sqrt{3}\bar{f}_{n+\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 - 51\sqrt{7}q^4 \\ + 13\sqrt{3}q^3 + 170\sqrt{7}q^2 - 119\sqrt{7}) \\ + 9\sqrt{3}\bar{f}_{n-\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 + 51\sqrt{7}q^4 + 13\sqrt{3}q^3 \\ - 170\sqrt{7}q^2 + 119\sqrt{7}) \left. \right] \left. \right],$$

$$\bar{f}_{n-q} = f(r_{n-q}, \bar{y}_{n-q})$$

The final scheme, $M_8(q)$, consists of the above approximations and the scheme

$$y_{n+1} - 2y_n + y_{n-1} = h^2 \left[t_0(f_{n+1} + f_{n-1}) + t_1f_n \right. \\ \left. + t_2 \left(\bar{f}_{n+\frac{\sqrt{21}}{3}} + \bar{f}_{n-\frac{\sqrt{21}}{3}} \right) + t_3(\bar{f}_{n+q} + \bar{f}_{n-q}) \right] \quad (151)$$

with

$$t_0 = \frac{406q^2 - 151}{3360(q^2 - 1)}, \quad t_1 = \frac{2324q^2 - 255}{2940q^2}, \\ t_2 = \frac{9(42q^2 - 13)}{7840(7 - 3q^2)}, \quad (152) \\ t_3 = \frac{17}{56q^2(q^2 - 1)(3q^2 - 7)}$$

The free parameter is defined in order to minimize the phase-lag. As a result of the above, the method where

$$q = \sqrt{\frac{1223 - \sqrt{653209}}{1530}}$$

has phase-lag of order ten and an interval of periodicity equal to (0, 28.7979).

Simos (1997f) has considered the following three parameter explicit eighth-order hybrid family of methods

$$\bar{y}_{n-q} = \frac{1}{171360} [1680qy_{n+1}(9q^7 - 56q^5 + 98q^3 - 51) \\ - 3360y_n(9q^8 - 56q^6 + 98q^4 - 51) \\ + 1680qy_{n-1}(9q^7 - 56q^5 + 98q^3 + 51)$$

$$\bar{y}_{n+1} = 2y_n - y_{n-1} + h^2 f_n$$

$$\bar{f}_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{12} (\bar{f}_{n+1} + 10f_n + f_{n-1})$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{104} (5\bar{f}_{n+1} + 146y_n - 47y_{n-1}) + \frac{h^2}{4992} (-59\bar{f}_{n+1} + 1438f_n + 253f_{n-1})$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{52} (3\bar{y}_{n+1} + 20y_n + 29y_{n-1}) + \frac{h^2}{4992} (41\bar{f}_{n+1} - 682f_n - 271f_{n-1})$$

$$\hat{\hat{y}}_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{60} \left[(\bar{f}_{n+1} + f_{n-1}) + 26f_n + 16 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \right]$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{128} (-25\hat{\hat{y}}_{n+1} + 205y_n - 15y_{n-1} - 37y_{n-2}) + \frac{h^2}{1536} (23\hat{\hat{y}}_{n+1} + 761f_n + 509f_{n-1} + 27f_{n-2})$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{128} [37(\hat{\hat{y}}_{n+1} + y_{n-2}) + 27(y_n + y_{n-1})] + \frac{h^2}{512} [-9(\hat{\hat{y}}_{n+1} + f_{n-2}) - 171(f_n + f_{n-1})]$$

$$\bar{y}_{n+\frac{1}{4}} = \frac{1}{4096} \left[605\hat{\hat{y}}_{n+1} + 4070y_n - 579y_{n-1} - 160 \left(\hat{y}_{n+\frac{1}{2}} - \hat{y}_{n-\frac{1}{2}} \right) \right] - \frac{h^2}{49152} \left[113\hat{\hat{y}}_{n+1} - 1390f_n - 103f_{n-1} + 1944 \left(\hat{f}_{n+\frac{1}{2}} - \hat{f}_{n-\frac{1}{2}} \right) \right]$$

$$\bar{y}_{n-\frac{1}{4}} = -\frac{1}{4096} \left[579\hat{\hat{y}}_{n+1} - 4070y_n - 605y_{n-1} - 160 \left(\hat{y}_{n+\frac{1}{2}} - \hat{y}_{n-\frac{1}{2}} \right) \right] + \frac{h^2}{49152} \left[103\hat{\hat{y}}_{n+1} + 1390f_n - 113f_{n-1} + 1944 \left(\hat{f}_{n+\frac{1}{2}} - \hat{f}_{n-\frac{1}{2}} \right) \right]$$

$$\bar{y}_n^{(i)} = y_n - a_i h^2 \left[\hat{\hat{f}}_{n+1} - 90f_n + f_{n-1} - 20 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) + 64 \left(\bar{f}_{n+\frac{1}{4}} + \bar{f}_{n-\frac{1}{4}} \right) \right],$$

$$i = 4(-1)0$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{3780} \left[47 \left(\hat{\hat{f}}_{n+1} + f_{n-1} \right) + 1328 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) - 1024 \left(\bar{f}_{n+\frac{1}{4}} + \bar{f}_{n-\frac{1}{4}} \right) + 3078\bar{f}_n^{(4)} \right] \quad (153)$$

The free parameters $a_i, i = 1(1)3$ are defined in order to minimize the phase-lag. As a result of the above, the method where $a_1 = 0$, $a_2 = -\frac{3644}{900315}$ and $a_3 = -\frac{198943}{211042260}$ has phase-lag of order twelve and an interval of periodicity equal to $(0, 12.9394)$. Also, the method where $a_1 = -\frac{122158423}{117312708240}$, $a_2 = -\frac{198943}{211042260}$ and $a_3 = -\frac{3644}{900315}$ has phase-lag of order fourteen and an interval of periodicity equal to $(0, 12.6756)$.

Simos (1997g) has considered the following two parameter explicit eighth-order hybrid method

$$\bar{y}_{n+1} = 2y_n - y_{n-1} + h^2 f_n$$

$$\bar{f}_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{12} (\bar{f}_{n+1} + 10f_n + f_{n-1})$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{104} (5\bar{f}_{n+1} + 146y_n - 47y_{n-1}) + \frac{h^2}{4992} (-59\bar{f}_{n+1} + 1438f_n + 253f_{n-1})$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{52} \left(3\bar{\bar{y}}_{n+1} + 20y_n + 29y_{n-1} \right) + \frac{h^2}{4992} \left(41\bar{\bar{f}}_{n+1} - 682f_n - 271f_{n-1} \right)$$

$$\hat{y}_n = y_n - ah^2 \left[\bar{\bar{f}}_{n+1} - 2f_n + f_{n-1} - 4 \left(\hat{f}_{n+\frac{1}{2}} - 2f_n + \hat{f}_{n-\frac{1}{2}} \right) \right]$$

$$\hat{\hat{y}}_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{60} \left[\left(\bar{\bar{f}}_{n+1} + f_{n-1} \right) + 26\hat{f}_n + 16 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \right]$$

$$\bar{y}_{n+s} = \frac{1}{486} \left[\sqrt{7} \{ \hat{y}_{n+1} (7\sqrt{7} + 27\sqrt{3}) + 192y_n + 3\sqrt{7}y_{n-1} (7\sqrt{7} - 27\sqrt{3}) + h^2 [2\sqrt{7}\bar{\bar{f}}_{n+1} (9\sqrt{3} + 7\sqrt{7}) + 224f_n + 2\sqrt{7}f_{n-1} (7\sqrt{7} - 9\sqrt{3})] \right],$$

$$s = \frac{\sqrt{21}}{3}$$

$$\bar{y}_{n-s} = \frac{1}{486} \left[\sqrt{7} \{ \hat{\hat{y}}_{n+1} (7\sqrt{7} - 27\sqrt{3}) + 192y_n + 3\sqrt{7}y_{n-1} (7\sqrt{7} + 27\sqrt{3}) + h^2 [2\sqrt{7}\bar{\bar{f}}_{n+1} (9\sqrt{3} - 7\sqrt{7}) + 224f_n + 2\sqrt{7}f_{n-1} (7\sqrt{7} + 9\sqrt{3})] \right],$$

$$s = \frac{\sqrt{21}}{3}$$

$$\bar{y}_{n+q} = \frac{1}{171360} \left[1680q\hat{\hat{y}}_{n+1} (9q^7 - 56q^5 + 98q^3 + 51) - 3360y_n (9q^8 - 56q^6 + 98q^4 - 51) + 1680qy_{n-1} (9q^7 - 56q^5 + 98q^3 - 51) - h^2q \left[7\bar{\bar{f}}_{n+1} (261q^7 - 1318q^5 + 459q^4 + 1057q^3 - 3570q^2 + 3111) + 16qf_n (747q^6 - 4801q^4 + 9409q^2 - 5355) + 7f_{n-1} (261q^7 - 1318q^5 - 459q^4 + 1057q^3 + 3570q^2 - 3111) + 9\sqrt{3}\bar{f}_{n+\frac{\sqrt{21}}{3}} (9\sqrt{3}q^7 - 22\sqrt{3}q^5 + 51\sqrt{7}q^4 + 13\sqrt{3}q^3 - 170\sqrt{7}q^2 + 119\sqrt{7}) + 9\sqrt{3}\bar{f}_{n-\frac{\sqrt{21}}{3}} (9\sqrt{3}q^7 - 22\sqrt{3}q^5 - 51\sqrt{7}q^4 + 13\sqrt{3}q^3 + 170\sqrt{7}q^2 - 119\sqrt{7}) \right] \right],$$

$$\bar{y}_{n-q} = \frac{1}{171360} \left[1680q\hat{\hat{y}}_{n+1} (9q^7 - 56q^5 + 98q^3 - 51) - 3360y_n (9q^8 - 56q^6 + 98q^4 - 51) + 1680qy_{n-1} (9q^7 - 56q^5 + 98q^3 + 51) - h^2q \left[7\bar{\bar{f}}_{n+1} (261q^7 - 1318q^5 - 459q^4 + 1057q^3 + 3570q^2 - 3111) + 16qf_n (747q^6 - 4801q^4 + 9409q^2 - 5355) + 7f_{n-1} (261q^7 - 1318q^5 + 459q^4 + 1057q^3 - 3570q^2 + 3111) + 9\sqrt{3}\bar{f}_{n+\frac{\sqrt{21}}{3}} (9\sqrt{3}q^7 - 22\sqrt{3}q^5 - 51\sqrt{7}q^4 + 13\sqrt{3}q^3 + 170\sqrt{7}q^2 - 119\sqrt{7}) + 9\sqrt{3}\bar{f}_{n-\frac{\sqrt{21}}{3}} (9\sqrt{3}q^7 - 22\sqrt{3}q^5 + 51\sqrt{7}q^4 + 13\sqrt{3}q^3 - 170\sqrt{7}q^2 + 119\sqrt{7}) \right] \right],$$

$$\begin{aligned}
& y_{n-1} - 2y_n + y_{n+1} \\
& = h^2 \left[t_0(f_{n+1} + f_{n-1}) + t_1 f_n + t_2 \right. \\
& \quad \left. \left(\bar{f}_{n+\frac{1}{2}} \frac{\sqrt{21}}{3} + \bar{f}_{n-\frac{1}{2}} \frac{\sqrt{21}}{3} \right) + t_3 (\bar{f}_{n+q} + f_{n-q}) \right]
\end{aligned} \quad (154)$$

where t_i , $i = 0(1)3$ are defined in Eq. (152).

The free parameters q and a are defined in order to minimize the phase-lag. As a result of the above, the method where $q = -2.39883862783148340941791501263$ and

$$\begin{aligned}
a &= 0.000561300458274047515866898432037\sqrt{7}\sqrt{3} \\
&+ 0.0124863707411472501777480876286
\end{aligned}$$

has phase-lag of order twelve and an interval of periodicity equal to (0, 14.4576).

Simos (1997h) has considered the following five parameter family of hybrid eighth-order methods

$$\begin{aligned}
\hat{y}_{n+\frac{1}{2}} &= \frac{1}{128}(-25y_{n+1} + 205y_n - 15y_{n-1} - 37y_{n-2}) \\
&+ \frac{h^2}{1536}(23f_{n+1} + 761f_n + 509f_{n-1} + 27f_{n-2})
\end{aligned}$$

$$\begin{aligned}
\hat{y}_{n-\frac{1}{2}} &= \frac{1}{128}[37(y_{n+1} + y_{n-2}) + 27(y_n + y_{n-1})] \\
&+ \frac{h^2}{512}[-9(f_{n+1} + f_{n-2}) - 171(f_n + f_{n-1})]
\end{aligned}$$

$$\begin{aligned}
\bar{y}_{n+\frac{1}{4}} &= \frac{1}{4096} \left[605y_{n+1} + 4070y_n - 579y_{n-1} \right. \\
&- 160 \left(\hat{y}_{n+\frac{1}{2}} - \hat{y}_{n-\frac{1}{2}} \right) \left. \right] - \frac{h^2}{49152} \left[113f_{n+1} - 1390f_n \right. \\
&- 103f_{n-1} + 1944 \left(\hat{f}_{n+\frac{1}{2}} - \hat{f}_{n-\frac{1}{2}} \right) \left. \right]
\end{aligned}$$

$$\begin{aligned}
\bar{y}_{n-\frac{1}{4}} &= -\frac{1}{4096} \left[579y_{n+1} - 4070y_n - 605y_{n-1} \right. \\
&- 160 \left(\hat{y}_{n+\frac{1}{2}} - \hat{y}_{n-\frac{1}{2}} \right) \left. \right] + \frac{h^2}{49152} \left[103f_{n+1} \right. \\
&+ 1390f_n - 113f_{n-1} + 1944 \left(\hat{f}_{n+\frac{1}{2}} - \hat{f}_{n-\frac{1}{2}} \right) \left. \right]
\end{aligned}$$

$$\begin{aligned}
\bar{y}_n^{(i)} &= y_n - a_i h^2 \left[f_{n+1} - 90f_n + f_{n-1} - 20 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \right. \\
&+ 64 \left(\bar{f}_{n+\frac{1}{4}} + \bar{f}_{n-\frac{1}{4}} \right) \left. \right],
\end{aligned}$$

$$i = 4(-1)0$$

$$\begin{aligned}
y_{n+1} - 2y_n + y_{n-1} &= \frac{h^2}{3780} \left[47(f_{n+1} + f_{n-1}) \right. \\
&+ 1328 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \\
&- 1024 \left(\bar{f}_{n+\frac{1}{4}} + \bar{f}_{n-\frac{1}{4}} \right) + 3078\bar{f}_n \left. \right]
\end{aligned} \quad (155)$$

The free parameters a_i , $i = 4(-1)0$ are defined in order to minimize the phase-lag. As a result of the above,

1. the method where $a_i = 0$, $i = 0(1)3$ and $a_4 = -\frac{236}{484785}$ has phase-lag of order ten and an interval of periodicity equal to (0, 29.1895).
2. the method where $a_i = 0$, $i = 0(1)2$, $a_3 = -\frac{233}{1471932}$ and $a_4 = -\frac{236}{484785}$ has phase-lag of order twelve and an interval of periodicity equal to (0, 32.3681).
3. the method where $a_i = 0$, $i = 0, 1$, $a_2 = -\frac{451037}{3434886000}$, $a_3 = -\frac{233}{1471932}$ and $a_4 = -\frac{236}{484785}$ has phase-lag of order fourteen and an interval of periodicity equal to (0, 34.9295).
4. the method where $a_0 = 0$, $a_1 = -\frac{3851}{35274204}$, $a_2 = -\frac{451037}{3434886000}$, $a_3 = -\frac{233}{1471932}$ and $a_4 = -\frac{236}{484785}$ has phase-lag of order sixteen and an interval of periodicity equal to (0, 37.0284).

5. the method where $a_0 = -\frac{1010455379}{11841212691000}$, $a_1 = -\frac{3851}{35274204}$, $a_2 = -\frac{451037}{3434886000}$, $a_3 = -\frac{233}{1471932}$ and $a_4 = -\frac{236}{484785}$ has phase-lag of order eighteen and an interval of periodicity equal to (0, 72.8593).

Simos (1997i) has considered the following two and three parameter families of hybrid eighth-order methods:

7.3.1. Two parameter family.

$$\bar{y}_{n+s} = \frac{1}{486} \left[\sqrt{7}y_{n+1}(7\sqrt{7} + 27\sqrt{3}) + 192y_n + 3\sqrt{7}y_{n-1}(7\sqrt{7} - 27\sqrt{3}) + h^2 \left[2\sqrt{7}f_{n+1}(9\sqrt{3} + 7\sqrt{7}) + 224f_n + 2\sqrt{7}f_{n-1}(7\sqrt{7} - 9\sqrt{3}) \right] \right], \quad s = \frac{\sqrt{21}}{3}$$

$$\bar{y}_{n+s} = \frac{1}{486} \left[\sqrt{7}y_{n+1}(7\sqrt{7} + 27\sqrt{3}) + 192y_n + 3\sqrt{7}y_{n-1}(7\sqrt{7} - 27\sqrt{3}) + h^2 \left[2\sqrt{7}f_{n+1}(9\sqrt{3} + 7\sqrt{7}) + 224f_n + 2\sqrt{7}f_{n-1}(7\sqrt{7} - 9\sqrt{3}) \right] \right], \quad s = \frac{\sqrt{21}}{3}$$

$$\begin{aligned} \bar{y}_{n+q} = \frac{1}{171360} & \left[1680qy_{n+1}(9q^7 - 56q^5 + 98q^3 + 51) \right. \\ & - 3360y_n(9q^8 - 56q^6 + 98q^4 - 51) \\ & + 1680qy_{n-1}(9q^7 - 56q^5 + 98q^3 - 51) \\ & - h^2q \left[7f_{n+1}(261q^7 - 1318q^5 + 459q^4 + 1057q^3 \right. \\ & - 3570q^2 + 3111) + 16qf_n(747q^6 - 4801q^4 \\ & + 9409q^2 - 5355) + 7f_{n-1}(261q^7 - 1318q^5 \\ & - 459q^4 + 1057q^3 + 3570q^2 - 3111) \\ & + 9\sqrt{3}\bar{f}_{n+\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 + 51\sqrt{7}q^4 \\ & + 13\sqrt{3}q^3 - 170\sqrt{7}q^2 + 119\sqrt{7}) \\ & + 9\sqrt{3}\bar{f}_{n-\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 - 51\sqrt{7}q^4 \\ & \left. \left. + 13\sqrt{3}q^3 + 170\sqrt{7}q^2 - 119\sqrt{7}) \right] \right], \end{aligned}$$

$$\begin{aligned} \bar{y}_{n-q} = \frac{1}{171360} & \left[1680qy_{n+1}(9q^7 - 56q^5 + 98q^3 - 51) \right. \\ & - 3360y_n(9q^8 - 56q^6 + 98q^4 - 51) \\ & + 1680qy_{n-1}(9q^7 - 56q^5 + 98q^3 + 51) \\ & - h^2q \left[7f_{n+1}(261q^7 - 1318q^5 - 459q^4 + 1057q^3 \right. \\ & + 3570q^2 - 3111) + 16qf_n(747q^6 - 4801q^4 \\ & + 9409q^2 - 5355) + 7f_{n-1}(261q^7 - 1318q^5 \\ & + 459q^4 + 1057q^3 - 3570q^2 + 3111) \\ & + 9\sqrt{3}\bar{f}_{n+\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 - 51\sqrt{7}q^4 \\ & + 13\sqrt{3}q^3 + 170\sqrt{7}q^2 - 119\sqrt{7}) \\ & + 9\sqrt{3}\bar{f}_{n-\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 + 51\sqrt{7}q^4 \\ & \left. \left. + 13\sqrt{3}q^3 - 170\sqrt{7}q^2 + 119\sqrt{7}) \right] \right], \end{aligned}$$

$$\begin{aligned} \bar{y}_n = y_n + w_0h^2 & \left[f_{n+1} - 2f_n + f_{n-1} \right. \\ & + z_2 \left(\bar{f}_{n+\frac{\sqrt{21}}{3}} + \bar{f}_{n-\frac{\sqrt{21}}{3}} - 2f_n \right) \\ & \left. + z_3 \left(\bar{f}_{n+q} + \bar{f}_{n-q} \right) - 2f_n \right] \\ y_{n+1} - 2y_n + y_{n-1} = h^2 & \left[t_0(f_{n+1} + f_{n-1}) + t_1\bar{f}_n \right. \\ & + t_2 \left(\bar{f}_{n+\frac{\sqrt{21}}{3}} + \bar{f}_{n-\frac{\sqrt{21}}{3}} \right) \\ & \left. + t_3 \left(\bar{f}_{n+q} + \bar{f}_{n-q} \right) \right] \quad (156) \end{aligned}$$

where t_i , $i = 0(1)3$ are defined in Eq. (152) and $z_2 = \frac{9(1-q^2)}{7(3q^2-7)}$ and $z_3 = \frac{4}{q^2(3q^2-7)}$.

The free parameters are defined in order to minimize the phase-lag. As a result of the above, the method where $w_0 = 0.0012658825222778819$ and $q = 1.24832627500665355549$, has phase-lag of order twelve and an interval of periodicity equal to (0, 32.5779).

7.3.2. Three parameter family.

$$\bar{y}_{n+s} = \frac{1}{486} \left[\sqrt{7}y_{n+1}(7\sqrt{7} + 27\sqrt{3}) + 192y_n \right. \\ \left. + 3\sqrt{7}y_{n-1}(7\sqrt{7} - 27\sqrt{3}) \right. \\ \left. + h^2 \left[2\sqrt{7}f_{n+1}(9\sqrt{3} + 7\sqrt{7}) + 224f_n \right. \right. \\ \left. \left. + 2\sqrt{7}f_{n-1}(7\sqrt{7} - 9\sqrt{3}) \right] \right], \quad s = \frac{\sqrt{21}}{3}$$

$$\bar{y}_{n-s} = \frac{1}{486} \left[\sqrt{7}y_{n+1}(7\sqrt{7} - 27\sqrt{3}) + 192y_n \right. \\ \left. + 3\sqrt{7}y_{n-1}(7\sqrt{7} + 27\sqrt{3}) \right. \\ \left. + h^2 \left[2\sqrt{7}f_{n+1}(9\sqrt{3} - 7\sqrt{7}) \right. \right. \\ \left. \left. + 224f_n + 2\sqrt{7}f_{n-1}(7\sqrt{7} + 9\sqrt{3}) \right] \right], \quad s = \frac{\sqrt{21}}{3}$$

$$\bar{y}_{n+q} = \frac{1}{171360} \left[1680qy_{n+1}(9q^7 - 56q^5 + 98q^3 + 51) \right. \\ - 3360y_n(9q^8 - 56q^6 + 98q^4 - 51) \\ + 1680qy_{n-1}(9q^7 - 56q^5 + 98q^3 - 51) \\ - h^2q \left[7f_{n+1}(261q^7 - 1318q^5 + 459q^4 + 1057q^3 \right. \\ - 3570q^2 + 3111) + 16qf_n(747q^6 - 4801q^4 \\ + 9409q^2 - 5355) + 7f_{n-1}(261q^7 - 1318q^5 \\ - 459q^4 + 1057q^3 + 3570q^2 - 3111) \\ + 9\sqrt{3}\bar{f}_{n+\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 + 51\sqrt{7}q^4 \\ + 13\sqrt{3}q^3 - 170\sqrt{7}q^2 + 119\sqrt{7}) \\ + 9\sqrt{3}\bar{f}_{n-\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 - 51\sqrt{7}q^4 \\ \left. \left. + 13\sqrt{3}q^3 + 170\sqrt{7}q^2 - 119\sqrt{7}) \right] \right],$$

$$\bar{y}_{n-q} = \frac{1}{171360} \left[1680qy_{n+1}(9q^7 - 56q^5 + 98q^3 - 51) \right. \\ - 3360y_n(9q^8 - 56q^6 + 98q^4 - 51) \\ + 1680qy_{n-1}(9q^7 - 56q^5 + 98q^3 + 51) \\ - h^2q \left[7f_{n+1}(261q^7 - 1318q^5 - 459q^4 + 1057q^3 \right. \\ \left. \left. + 3570q^2 - 3111) + 16qf_n(747q^6 - 4801q^4) \right. \right. \\ \left. \left. + 7f_{n-1}(261q^7 - 1318q^5 + 459q^4 + 1057q^3 \right. \right. \\ \left. \left. + 3570q^2 - 5355) + 7f_{n-1}(261q^7 - 1318q^5 \right. \right. \\ \left. \left. - 459q^4 + 1057q^3 + 3570q^2 - 3111) \right. \right. \\ \left. \left. + 9\sqrt{3}\bar{f}_{n+\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 + 51\sqrt{7}q^4 \right. \right. \\ \left. \left. + 13\sqrt{3}q^3 - 170\sqrt{7}q^2 + 119\sqrt{7}) \right. \right. \\ \left. \left. + 9\sqrt{3}\bar{f}_{n-\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 - 51\sqrt{7}q^4 \right. \right. \\ \left. \left. + 13\sqrt{3}q^3 + 170\sqrt{7}q^2 - 119\sqrt{7}) \right] \right],$$

$$+ 9409q^2 - 5355) + 7f_{n-1}(261q^7 - 1318q^5 \\ + 459q^4 + 1057q^3 - 3570q^2 + 3111) \\ + 9\sqrt{3}\bar{f}_{n+\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 - 51\sqrt{7}q^4 \\ + 13\sqrt{3}q^3 + 170\sqrt{7}q^2 - 119\sqrt{7}) \\ + 9\sqrt{3}\bar{f}_{n-\frac{\sqrt{21}}{3}}(9\sqrt{3}q^7 - 22\sqrt{3}q^5 + 51\sqrt{7}q^4 \\ + 13\sqrt{3}q^3 - 170\sqrt{7}q^2 + 119\sqrt{7}) \left. \right] \right],$$

$$\bar{y}_n = y_n + w_0 h^2 \left[f_{n+1} - 2f_n + f_{n-1} \right. \\ \left. - \frac{3}{7} \left(\bar{f}_{n+\frac{\sqrt{21}}{3}} + \bar{f}_{n-\frac{\sqrt{21}}{3}} - 2f_n \right) \right]$$

$$\hat{y}_n = y_n + w_1 h^2 \left[f_{n+1} - 2\bar{f}_n + f_{n-1} \right. \\ \left. + z_2 \left(\bar{f}_{n+\frac{\sqrt{21}}{3}} + \bar{f}_{n-\frac{\sqrt{21}}{3}} - 2\bar{f}_n \right) \right. \\ \left. + z_3 \left(\bar{f}_{n+q} + \bar{f}_{n-q} \right) - 2f_n \right]$$

$$y_{n+1} - 2y_n + y_{n-1} = h^2 \left[t_0(f_{n+1} + f_{n-1}) + t_1\hat{f}_n \right. \\ \left. + t_2 \left(\bar{f}_{n+\frac{\sqrt{21}}{3}} + \bar{f}_{n-\frac{\sqrt{21}}{3}} \right) \right. \\ \left. + t_3 \left(\bar{f}_{n+q} + \bar{f}_{n-q} \right) \right] \quad (157)$$

where t_i , $i = 0(1)3$ are defined in Eq. (152) and z_2 , z_3 are defined above.

The free parameters are defined in order to minimize the phase-lag. As a result of the above, the method where $w_0 = 0.00864946221830391659$, $w_1 = 0.0018697764609313898$ and $q = 1.30364491397327910153$, has phase-lag of order fourteen and an interval of periodicity equal to $(0, 62.7458)$.

Simos and Touloulidis (1997) have considered the following one-parameter explicit eighth-order hybrid method:

$$\bar{y}_{n+1} = 2y_n - y_{n-1} + h^2 f_n$$

$$\bar{\bar{y}}_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{12} (\bar{f}_{n+1} + 10f_n + f_{n-1})$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{104} (5\bar{\bar{y}}_{n+1} + 146y_n - 47y_{n-1}) + \frac{h^2}{4992} (-59\bar{\bar{f}}_{n+1} + 1438f_n + 253f_{n-1})$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{52} (3\bar{\bar{y}}_{n+1} + 20y_n + 29y_{n-1}) + \frac{h^2}{4992} (41\bar{\bar{f}}_{n+1} - 682f_n - 271f_{n-1})$$

$$\hat{\hat{y}}_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{60} \left[(\bar{\bar{f}}_{n+1} + f_{n-1}) + 26\hat{f}_n + 16 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \right]$$

$$\bar{y}_{n+s} = \frac{1}{486} \left[\sqrt{7}\hat{\hat{y}}_{n+1} (7\sqrt{7} + 27\sqrt{3}) + 192y_n + 3\sqrt{7}y_{n-1} (7\sqrt{7} - 27\sqrt{3}) + h^2 \left[2\sqrt{7}\bar{\bar{f}}_{n+1} (9\sqrt{3} + 7\sqrt{7}) + 224f_n + 2\sqrt{7}f_{n-1} (7\sqrt{7} - 9\sqrt{3}) \right] \right], \quad s = \frac{\sqrt{21}}{3}$$

$$\bar{y}_{n-s} = \frac{1}{486} \left[\sqrt{7}\hat{\hat{y}}_{n+1} (7\sqrt{7} - 27\sqrt{3}) + 192y_n + 3\sqrt{7}y_{n-1} (7\sqrt{7} + 27\sqrt{3}) + h^2 \left[2\sqrt{7}\bar{\bar{f}}_{n+1} (9\sqrt{3} - 7\sqrt{7}) + 224f_n + 2\sqrt{7}f_{n-1} (7\sqrt{7} + 9\sqrt{3}) \right] \right], \quad s = \frac{\sqrt{21}}{3}$$

$$\bar{y}_{n+q} = \frac{1}{171360} \left[1680q\hat{\hat{y}}_{n+1} (9q^7 - 56q^5 + 98q^3 + 51) - 3360y_n (9q^8 - 56q^6 + 98q^4 - 51) + 1680qy_{n-1} (9q^7 - 56q^5 + 98q^3 - 51) - h^2q \left[7\bar{\bar{f}}_{n+1} (261q^7 - 1318q^5 + 459q^4 + 1057q^3 - 3570q^2 + 3111) + 16qf_n (747q^6 - 4801q^4 + 9409q^2 - 5355) + 7f_{n-1} (261q^7 - 1318q^5 - 459q^4 + 1057q^3 + 3570q^2 - 3111) + 9\sqrt{3}\bar{f}_{n+\frac{1}{2}} (9\sqrt{3}q^7 - 22\sqrt{3}q^5 + 51\sqrt{7}q^4 + 13\sqrt{3}q^3 - 170\sqrt{7}q^2 + 119\sqrt{7}) + 9\sqrt{3}\bar{f}_{n-\frac{1}{2}} (9\sqrt{3}q^7 - 22\sqrt{3}q^5 - 51\sqrt{7}q^4 + 13\sqrt{3}q^3 + 170\sqrt{7}q^2 - 119\sqrt{7}) \right] \right],$$

$$\bar{y}_{n-q} = \frac{1}{171360} \left[1680q\hat{\hat{y}}_{n+1} (9q^7 - 56q^5 + 98q^3 - 51) - 3360y_n (9q^8 - 56q^6 + 98q^4 - 51) + 1680qy_{n-1} (9q^7 - 56q^5 + 98q^3 + 51) - h^2q \left[7\bar{\bar{f}}_{n+1} (261q^7 - 1318q^5 - 459q^4 + 1057q^3 - 3570q^2 - 3111) + 16qf_n (747q^6 - 4801q^4 + 9409q^2 - 5355) + 7f_{n-1} (261q^7 - 1318q^5 + 459q^4 + 1057q^3 - 3570q^2 + 3111) + 9\sqrt{3}\bar{f}_{n+\frac{1}{2}} (9\sqrt{3}q^7 - 22\sqrt{3}q^5 - 51\sqrt{7}q^4 + 13\sqrt{3}q^3 + 170\sqrt{7}q^2 - 119\sqrt{7}) + 9\sqrt{3}\bar{f}_{n-\frac{1}{2}} (9\sqrt{3}q^7 - 22\sqrt{3}q^5 + 51\sqrt{7}q^4 + 13\sqrt{3}q^3 - 170\sqrt{7}q^2 + 119\sqrt{7}) \right] \right],$$

$$\begin{aligned}
y_{n+1} - 2y_n + y_{n-1} = & h^2 \left[t_0(f_{n+1} + f_{n-1}) + t_1 f_n \right. \\
& + t_2 \left(\bar{f}_{n+\frac{1}{2}} + \bar{f}_{n-\frac{1}{2}} \right) \\
& \left. + t_3 (\bar{f}_{n+q} + \bar{f}_{n-q}) \right] \quad (158)
\end{aligned}$$

where t_i , $i = 0(1)3$ are defined in Eq. (152).

The free parameter q is defined in order to minimize the phase-lag. As a result of the above, the method where

$$q = \sqrt{\frac{661 - \sqrt{338137}}{882}},$$

has phase-lag of order ten and an interval of periodicity equal to (0, 9.5301).

7.4. Variable-step methods

Simos (1993f) has considered the following family of methods:

$$\bar{y}_{n+1} = 2y_n - y_{n-1} + h^2 f_n$$

$$\bar{y}_n = y_n - ah^2(\bar{f}_{n+1} - 2f_n + f_{n-1})$$

$$\bar{\bar{y}}_n = y_n - bh^2(\bar{\bar{f}}_{n+1} - 2\bar{f}_n + f_{n-1})$$

$$\hat{y}_n = y_n - ch^2(\hat{f}_{n+1} - 2\bar{\bar{f}}_n + f_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12} \left[(\bar{f}_{n+1} + f_{n-1}) + 10\hat{f}_n \right] \quad (159)$$

The value of parameters a , b and c are determined in order to satisfy the minimal phase-lag property. As a result of the above, a method with phase-lag of order ten and with an interval of periodicity equal to (0, 31.70) is produced.

Based on the above method and the method (Simos, 1991c)) a variable-step procedure is developed.

Simos (1993g) has considered the following families of hybrid sixth algebraic order methods:

7.4.1. Two parameter family

$$\bar{y}_n = y_n - ah^2(f_{n+1} - 2f_n + f_{n-1})$$

$$\hat{y}_n = y_n - bh^2(f_{n+1} - 2\bar{f}_n + f_{n-1})$$

$$\hat{\bar{y}}_{n+\frac{1}{2}} = \frac{1}{2}(y_{n+1} + \hat{y}_n) - h^2 \left[\left(\frac{1}{8} - c \right) f_{n+1} + c\hat{f}_n \right]$$

$$\hat{\bar{y}}_{n-\frac{1}{2}} = \frac{1}{2}(y_{n-1} + \hat{y}_n) - h^2 \left[\left(\frac{1}{8} - c \right) f_{n-1} + c\hat{f}_n \right]$$

$$\hat{\hat{y}}_{n+\frac{1}{2}} = \frac{1}{2}(y_{n+1} + \bar{y}_n) - \frac{h^2}{96} \left(f_{n+1} + 10\hat{\bar{f}}_{n+\frac{1}{2}} + f_n \right)$$

$$\hat{\hat{y}}_{n-\frac{1}{2}} = \frac{1}{2}(y_{n-1} + \bar{y}_n) - \frac{h^2}{96} \left(f_{n-1} + 10\hat{\bar{f}}_{n-\frac{1}{2}} + f_n \right)$$

$$\begin{aligned}
y_{n+1} - 2y_n + y_{n-1} = & \frac{h^2}{60} \left[(f_{n+1} + f_{n-1}) + 26f_n \right. \\
& \left. + 16 \left(\hat{\bar{f}}_{n+\frac{1}{2}} + \hat{\bar{f}}_{n-\frac{1}{2}} \right) \right] \quad (160)
\end{aligned}$$

The value of parameters a , b , and c are determined in order to satisfy the minimal phase-lag P-stability properties. As a result of the above, all the methods of the family with $c < -0.0191$ are P-stable methods with phase-lag of order ten.

7.4.2. Three parameter family

$$\bar{y}_n = y_n - ah^2(f_{n+1} - 2f_n + f_{n-1})$$

$$\hat{y}_n = y_n - bh^2(f_{n+1} - 2\bar{f}_n + f_{n-1})$$

$$\hat{\bar{y}}_n = y_n - ch^2(f_{n+1} - 2\hat{f}_n + f_{n-1})$$

$$\hat{\bar{y}}_{n+\frac{1}{2}} = \frac{1}{2}(y_{n+1} + \hat{y}_n) - h^2 \left[\left(\frac{1}{8} - d \right) f_{n+1} + d\hat{f}_n \right]$$

$$\hat{\bar{y}}_{n-\frac{1}{2}} = \frac{1}{2}(y_{n-1} + \hat{y}_n) - h^2 \left[\left(\frac{1}{8} - d \right) f_{n-1} + d\hat{f}_n \right]$$

$$\hat{\hat{y}}_{n+\frac{1}{2}} = \frac{1}{2}(y_{n+1} + \bar{y}_n) - \frac{h^2}{96} \left(f_{n+1} + 10\hat{\bar{f}}_{n+\frac{1}{2}} + f_n \right)$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{2}(y_{n-1} + \bar{y}_n) - \frac{h^2}{96} \left(f_{n-1} + 10\hat{f}_{n-\frac{1}{2}} + f_n \right)$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{60} \left[(f_{n+1} + f_{n-1}) + 26f_n \right. \\ \left. + 16 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \right] \quad (161)$$

The value of parameters a , b , c and d are determined in order to satisfy the minimal phase-lag P-stability properties. As a result of the above, all the methods of the family with $a < -0.0239$ are P-stable methods with phase-lag of order twelve.

Based on the above, methods a variable-step procedure is developed. Simos (1994b) has considered the following families of sixth algebraic order methods:

7.4.3. One-parameter family

$$\bar{y}_{n+1} = y_{n+1} + ah^2(f_n - f_{n+1})$$

$$\bar{y}_{n-1} = y_{n-1} + ah^2(f_n - f_{n-1})$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{104}(5y_{n+1} + 146y_n - 47y_{n-1}) \\ + \frac{h^2}{4992}(-59\bar{f}_{n+1} + 1438f_n + 253f_{n-1})$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{52}(3y_{n+1} + 20y_n + 29y_{n-1}) \\ + \frac{h^2}{4992}(41\bar{f}_{n+1} - 682f_n - 271\bar{f}_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{60} \left[(f_{n+1} + f_{n-1}) + 26f_n \right. \\ \left. + 16 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \right] \quad (162)$$

The value of parameter a is determined in order to satisfy the minimal phase-lag property. As a result of the above, a method with phase-lag of order eight and with an interval of periodicity equal to $(0, 26)$ is produced.

7.4.4. Two-parameters family

$$\bar{y}_{n+1} = y_{n+1} + ah^2(f_n - f_{n+1})$$

$$\bar{y}_{n-1} = y_{n-1} + ah^2(f_n - f_{n-1})$$

$$\bar{y}_n = y_n - bh^2(\bar{f}_{n+1} - 2f_n + \bar{f}_{n-1})$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{104}(5y_{n+1} + 146y_n - 47y_{n-1}) \\ + \frac{h^2}{4992}(-59f_{n+1} + 1438\hat{f}_n + 253f_{n-1})$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{52}(3y_{n+1} + 20y_n + 29y_{n-1}) \\ + \frac{h^2}{4992}(41f_{n+1} - 682\hat{f}_n - 271f_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{60} \left[(f_{n+1} + f_{n-1}) + 26f_n \right. \\ \left. + 16 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \right] \quad (163)$$

The value of parameters a and b are determined in order to satisfy the minimal phase-lag property. As a result of the above, a method with phase-lag of order ten and with an interval of periodicity equal to $(0, 26)$ is produced.

Based on the above methods, a variable-step procedure is developed. Avdelas and Simos (1996a) have considered the following embedded, *automatically* defined, families of methods:

7.4.5. Fourth-order embedded method

$$\bar{y}_{n+1} = 2y_n - y_{n-1} + h^2f_n$$

$$y_{n,k,b} = y_n - a_{k,b}h^2(\bar{f}_{n+1} - 2f_{n,k-1,b} + f_{n-1})$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{12} \left[(\bar{f}_{n+1} + f_{n-1}) + 10f_{n,b,b} \right] \quad (164)$$

where k is the number of layer and b is the number of family. Based on the theorems proved and described by Avdelas and Simos (1996a), the parameters of the

above method are determined automatically in order to satisfy the phase-lag property.

7.4.6. Sixth-order embedded method

$$\bar{y}_{n+1} = 2y_n - y_{n-1} + h^2 f_n$$

$$\bar{\bar{y}}_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{12} (\bar{f}_{n+1} + 10f_n + f_{n-1})$$

$$\hat{y}_{n+\frac{1}{2}} = \frac{1}{104} (5\bar{\bar{y}}_{n+1} + 146y_n - 47y_{n-1}) + \frac{h^2}{4992} (-59\bar{f}_{n+1} + 1438f_n + 253f_{n-1})$$

$$\hat{y}_{n-\frac{1}{2}} = \frac{1}{52} (3\bar{\bar{y}}_{n+1} + 20y_n + 29y_{n-1}) + \frac{h^2}{4992} (41\bar{f}_{n+1} - 682f_n - 271f_{n-1})$$

$$y_{n,k,b} = y_n - a_{k,b} h^2 \left[\bar{\bar{f}}_{n+1} - 2f_{n,k-1,b} + f_{n-1} - 4 \left(\hat{f}_{n+\frac{1}{2}} - 2f_{n,k-1,b} + \hat{f}_{n-\frac{1}{2}} \right) \right]$$

$$y_{n+1} - 2y_n + y_{n-1} = \frac{h^2}{60} \left[(\bar{\bar{f}}_{n+1} + f_{n-1}) + 26f_{n,b,b} + 16 \left(\hat{f}_{n+\frac{1}{2}} + \hat{f}_{n-\frac{1}{2}} \right) \right] \quad (165)$$

where, as above, k is the number of layer and b is the number of family. Based on the theorems fully proved and described by Avdelas and Simos (1996a), the parameters of the above method are determined *automatically* in order to satisfy the phase-lag property.

7.5. P-stable methods of high exponential order

Simos (1993h) has considered the following family of methods:

$$y_{n,m-j} = y_n - h^2 (b_{0,j+1} f_{n+1} + b_{1,j+1} f_{n,m-j-1} + b_{0,j+1} f_{n-1})$$

$$a_0 y_{n+1} + a_1 y_n + a_0 y_{n-1}$$

$$= h^2 (b_0 f_{n+1} + b_1 f_{n,m} + b_0 f_{n-1}) \quad (166)$$

Based on the theory fully described by Simos (1993h), two P-stable methods of exponential and phase-lag order eight and ten are produced.

For the above family of methods Avdelas and Simos (1996b) have given formulae for the coefficients of the P-stable methods in order to have exponential order and phase-lag order up to infinity. We note that the coefficients of the P-stable method, now, are determined *automatically*. This new approach is called *generator of methods*.

8. Methods based on transmission line theory

Following the usual procedure of the piecewise perturbation numerical method (Ixaru, 1984), the domain $[x_0, x_k]$ is divided by the mesh points

$$x_0, x_1, x_2, \dots, x_k \quad (167)$$

into k arbitrary intervals

$$[x_n, x_{n+1}], \quad n = 0, 1, 2, 3, \dots, k-1. \quad (168)$$

In addition, we denote $\Delta x_n = x_{n+1} - x_n$. Within each of these elementary intervals, we choose a polynomial $W_n(x)$, which approximates the true potential $W(x)$ over the interval. This piecewise-polynomial approximating curve is usually called a 'reference potential'.

The solutions for this problem satisfy the equation:

$$y''(x) + (E - W_n(x))y(x) = 0, \quad (169)$$

$$x_n \leq x \leq x_{n+1}, \quad n = 1, 2, 3, \dots, k \quad (170)$$

where $W_n(x)$ is a piecewise-polynomial reference potential and y and y' are continuous at the meshpoints x_n .

Eqs. (169) and (170) are equivalent to the following first-order differential equations (Papageorgiou and Raptis, 1987):

$$\partial I(x)/\partial x = -iV(x), \quad (171)$$

$$\partial V(x)/\partial x = -[(W_n(x) - E)/i]I(x) \quad (172)$$

where $W_n(x)$ is a *piecewise-polynomial reference potential* which replaces the *potential function* $W(x)$, in the sense that any solution of Eqs. (169) and (170) is a solution of the system and vice-versa.

We consider an intermediate point x_c and apply the recursion relations, which we obtain based on the piecewise perturbation numerical method. Starting from

x_0 outwards to x_c , we calculate $Z^L(x_c)$ (i.e., the *Ricatti variable to the left* (see Johnson, 1978) and starting from x_k inwards to x_c , we calculate $Z^R(x_c)$ (i.e., the *Ricatti variable to the right*). From (Johnson, 1978) we have that E is an eigenvalue if:

$$Z^L(x_c) + Z^R(x_c) = 0 \quad (173)$$

For a trial eigenvalue E , we have the value of the function

$$F(E) = Z^L(x_c) + Z^R(x_c) \quad (174)$$

To find the roots of the algebraic equation $F(E) = 0$ (which are the eigenvalues), we use the Secant method.

In order to calculate $Z^L(x_c)$ and $Z^R(x_c)$ for a given value E , the initial values $Z^L(x_0)$ and $Z^R(x_k)$ are required. These values are calculated from the equation

$$Z^L(0) = V(0)/I(0) = (\partial y / \partial x|_{x=0})/y(0) \quad (175)$$

and since, by definition, $y(0) = 0$ we have $Z^L(0) \rightarrow \infty$. Thus, for numerical purposes it is sufficient that $Z^L(x_0)$ be a very large number and in practice we take it equal to 10^{10} . Based on the analysis given by Papageorgiou et al. (1990a, b) and Johnson, 1978, we have that for large values of x we have $x_k \leq x \leq \infty$ and $W(x) \cong 0$ and $Z^R(x_k) = \sqrt{E} = -i\sqrt{|E|}$ (E is a negative number).

The piecewise perturbation numerical method which is used to approximate the function $Z(x) = y'(x)/y(x)$, where $y'(x) = dy(x)/dx$, at each step is described with full details by Ixaru (1984).

Based on this theory,

$$\begin{aligned} y(X+h)^H &= [u_0(h) + u_1(h) + u_2(h) \\ &+ \dots] y(X)^H + [v_0(h) + v_1(h) + v_2(h) \\ &+ \dots] y'(X)^H \end{aligned} \quad (176)$$

$$\begin{aligned} y'(X+h)^H &= [u'_0(h) + u'_1(h) + u'_2(h) \\ &+ \dots] y(X)^H + [v'_0(h) + v'_1(h) + v'_2(h) \\ &+ \dots] y'(X)^H \end{aligned} \quad (177)$$

where $u_0(h)$, $u_1(h)$, $u_2(h)$, $v_0(h)$, $v_1(h)$, $v_2(h)$, $u'_0(h)$, $u'_1(h)$, $u'_2(h)$, $v'_0(h)$, and $v'_1(h)$ are completely defined by Ixaru (1984). The subscript denotes the order of corrections. It is obvious that u_0 , for example, denotes the zero-order solution.

Simos (1995e) has developed a method based on the above theory and on the zero-order solution and first and second-order corrections.

Table 4

Modified five-stage Explicit Runge–Kutta method of order four

0					
1/2	1/2				
1/2	0	1/2			
1	0	0	1		
1	1/6	1/3	1/3	1/6	
	$c_{0,0}$	$c_{0,1}$	$c_{0,2}$	$c_{0,3}$	$c_{0,4}$

Simos and Tougelidis (1996b) have developed a variable-step method. Based on the above theory and on the zero-order solution, and first correction, an approximation of $y(X+h)$ and $y'(X+h)$ is obtained. Based on this approximation and on the method developed above (Simos, 1995e), a variable-step algorithm is produced.

Based on the Step Function Perturbative Numerical Method (SF-PNM) fully described by Ixaru (1984), Papageorgiou et al. (1990b) have considered the following formulae to calculate $y(X+h)$ and $y'(X+h)$:

$$\begin{aligned} y(X+h) &= \left[X_1 h - \frac{1}{2} C(h) z(h) \right] y(X) + h X_2(h) y'(X) \\ y'(X+h) &= w^2 h X_2(h) y(X) \\ &+ \left[X_1 h + \frac{1}{2} C(h) z(h) \right] y'(X) h X_2(h) y'(X) \end{aligned} \quad (178)$$

where

$$\begin{aligned} w &= \sqrt{|U_n - E|}, \quad X_1 = \cosh(wh), \quad X_2 = \sinh(wh)/wh, \quad C(h) \\ &= \left[U\left(X + \frac{h}{2}\right) - U\left(X - \frac{h}{2}\right) \right] h^2 \\ \text{and } z(h) &= \frac{[X_1(h) - X_2(h)]}{(2wh)^2} \end{aligned}$$

Table 5

Modified six-stage explicit runge–Kutta method of order four

0						
1/2	1/2					
1/2	0	1/2				
1	0	0	1			
1	1/6	1/3	1/3	1/6		
1	$c_{0,0}$	$c_{0,1}$	$c_{0,2}$	$c_{0,3}$	$c_{0,4}$	
	$c_{1,0}$	$c_{1,1}$	$c_{1,2}$	$c_{1,3}$	$c_{1,4}$	$c_{1,5}$

The above method has been applied to the eigenvalue problem of the Schrödinger equation. Papageorgiou et al. (1990a) have applied the above theory to the phase shift problem.

9. Runge–Kutta methods for specific Schrödinger equations

Simos and Williams (1996c) have considered the following Runge–Kutta methods, based on the well known classical fourth-order Runge–Kutta method Tables 4–6. The free parameters of the above methods are defined in order to minimize the phase-lag.

10. Numerical illustrations

In this section, we test several finite difference methods to the numerical solution of resonance and eigen-value problems of the Schrödinger equations to examine their efficiency. First, we examine the accuracy of exponentially fitted and phase-fitted methods. Next, we will examine the accuracy of some important methods with constant coefficients. Then we will examine the computational efficiency of some variable-step procedures. We note here that Bessel and Neumann fitted methods will be examined as a part of variable-step step procedures due to their large cost in a constant step procedure (it is known that the coefficients of the Bessel and Neumann fitted methods are radius dependent i.e., it is required to be recalculated at every step).

10.1. Exponentially fitted methods and phase fitted methods

We consider the numerical integration of the Schrödinger equation:

$$y''(x) = (W(x) - E)y(x) \quad (179)$$

in the well-known case where the potential $V(x)$ is the Woods–Saxon potential

$$V(x) = V_w(x) = \frac{u_0}{(1+z)} - \frac{u_0 z}{[a(1+z)^2]} \quad (180)$$

with $z = \exp[(x - X_0)/a]$, $u_0 = -50$, $a = 0.6$ and $X_0 = 7.0$. In order to solve this problem numerically we need to approximate the true (infinite) interval of integration $0, \infty$ by a finite interval. For the purpose of our numerical illustration we take the domain of integration as $0 \leq x \leq 15$. We consider Eq. (179) in a rather large domain of energies, i.e., $E \in [1, 1000]$. The problems we consider are (i) the so-called *resonance problem* and (ii) the so-called *bound-states problem*.

Table 6

Modified seven-stage explicit Runge–Kutta method of order four

0							
1/2	1/2						
1/2	0	1/2					
1/6	1/3	1/3	1/6				
$c_{0,0}$	$c_{0,1}$	$c_{0,2}$	$c_{0,3}$	$c_{0,4}$			
$c_{1,0}$	$c_{1,1}$	$c_{1,2}$	$c_{1,3}$	$c_{1,4}$	$c_{1,5}$		
	$c_{2,0}$	$c_{2,1}$	$c_{2,2}$	$c_{2,3}$	$c_{2,4}$	$c_{2,5}$	$c_{2,6}$

10.1.1. The resonance problem—Woods–Saxon potential

In the case of positive energies $E = k^2$ the potential dies away faster than the term $l(l+1)/x^2$ and the Schrödinger equation effectively reduces to

$$y''(x) + \left(k^2 - \frac{l(l+1)}{x^2}\right)y(x) = 0 \quad (181)$$

for x greater than some value X .

The above equation has linearly independent solutions $kxj_l(kx)$ and $kxn_l(kx)$, where $j_l(kx)$, $n_l(kx)$ are the *spherical Bessel functions* and *Neumann functions*, respectively. Thus the solution of Eq. (1) has (when $x \rightarrow 0$) the asymptotic form

$$y(x) \simeq Akxj_l(kx) - Bkxn_l(kx) \\ \simeq AC[\sin(kx - \pi l/2) + \tan \delta_l \cos(kx - \pi l/2)] \quad (182)$$

where δ_l is the *phase shift* that may be calculated from the formula

$$\tan \delta_l = \frac{y(x_2)S(x_1) - y(x_1)S(x_2)}{y(x_1)C(x_2) - y(x_2)C(x_1)} \quad (183)$$

for x_1 and x_2 distinct points on the asymptotic region (for which we have that x_1 is the right-hand end point of the interval of integration and $x_2 = x_1 - h$, h is the stepsize) with $S(x) = kxj_l(kx)$ and $C(x) = kxn_l(kx)$.

Since the problem is treated as an initial-value problem, we need y_0 and y_1 before starting a two-step method. From the initial condition, $y_0 = 0$. It can be shown that, for values of x close to the origin, the solution behaves like $y(x) \sim cx^{l+1}$ as $x \rightarrow 0$, where c is an independent constant. In view of this we take $y_1 = h^{l+1}$. With these starting values we evaluate at x_1 of the asymptotic region the phase shift δ_l and the normalization factor C from the above relations. If we want more starting values then Runge–Kutta–Nyström methods are used (Hairer et al., 1987).

For positive energies one has the so-called resonance problem. This problem consists either of finding the phase shift $\delta(E) = \delta_l$ or finding those E , for

Table 7

Maximum absolute errors, of the resonances calculated by the 26 algorithms MI–MXXVI for the resonance problem with the Woods–Saxon potential^a

Method	Maximum absolute error MI
MII	
MIII	
MIV	
MV	
MVI	7.2×10^{-2}
MVII	3.1×10^{-2}
MVIII	4.4×10^{-3}
MIX	3.5×10^{-4}
MIX	7.7×10^{-2}
MX	
MXI	
MXII	4.5×10^{-2}
MXIII	9.9×10^{-3}
MXIV	5.6×10^{-5}
MXV	4.1×10^{-6}
MXVI	
MXVII	
MXVIII	4.4×10^{-2}
MXIX	9.9×10^{-3}
MXX	5.4×10^{-5}
MXXI	
MXXII	
MXXIII	9.1×10^{-1}
MXXIV	4.2×10^{-1}
MXXV	5.6×10^{-1}
MXXVI	8.3×10^{-2}

^a The empty areas indicate that the maximum absolute error is greater than 1.

$E \in [1, 1000]$, at which δ equals $\pi/2$. We actually solve the latter problem, known as ‘the resonance problem’ when *the positive eigenenergies lie under the potential barrier*.

The boundary conditions for this problem are:

$$y(0) = 0,$$

$$y(x) \sim \cos[\sqrt{E}x] \text{ for large } x.$$

The domain of numerical integration is $[0, 15]$. For comparison purposes we use the following procedures:

Method MI	Numerov’s method
Method MII	Derived by Ixaru and Rizea (1980)
Method MIII	Derived by Raptis and Cash (1987)
Method MIV	Derived by Cash et al. (1990)
Method MV	Derived by Simos (1992a)
Method MVI	Method of Thomas et al. (1996) (Case I of the family)

Table 8

Maximum absolute errors, of the positive eigenvalues calculated by the 26 algorithms MI–MXXVI for the resonance problem with the Woods–Saxon potential^a

Method	Maximum absolute error MI
MII	
MIII	
MIV	
MV	
MVI	
MVII	
MVIII	
MIX	3.1×10^{-1}
MX	
MXI	8.9×10^{-1}
MXII	9.6×10^{-2}
MXIII	5.5×10^{-2}
MXIV	3.5×10^{-3}
MXV	4.1×10^{-4}
MXVI	
MXVII	8.9×10^{-1}
MXVIII	9.6×10^{-2}
MXIX	5.3×10^{-2}
MXX	3.4×10^{-3}
MXXI	
MXXII	
MXXIII	8.9×10^{-1}
MXXIV	4.1×10^{-1}
MXXV	5.3×10^{-1}
MXXVI	7.8×10^{-2}

^a The empty areas indicate that the maximum absolute error is greater than 1.

Method MVII	Method of Thomas et al. (1996) (Case II of the family)
Method MVIII	Method of Thomas et al. (1996) (Case III of the family)
Method MIX	Method of Thomas et al. (1996) (Case IV of the family)
Method MX	Method of Simos and Williams (1997a) (Case I of the family)
Method MXI	Method of Simos and Williams (1997a) (Case II of the family)
Method MXII	Method of Simos and Williams (1997a) (Case III of the family)
Method MXIII	Method of Simos and Williams (1997a) (Case IV of the family)
Method MXIV	Method of Simos and Williams (1997a) (Case V of the family)
Method MXV	Method of Simos and Williams (1997a) (Case VI of the family)
Method MXVI	Hybrid sixth-order method of Thomas and Simos (1997) (Case I of the family)

Method MXVII	Hybrid sixth-order method of Thomas and Simos (1997) (Case II of the family)
Method MXVIII	Hybrid sixth-order method of Thomas and Simos (1997) (Case III of the family)
Method MXIX	Hybrid sixth-order method of Thomas and Simos (1997) (Case IV of the family)
Method MXX	Hybrid sixth-order method of Thomas and Simos (1997) (Case V of the family)
Method MXXI	Sixth-order phase fitted method of Simos (1991b)
Method MXXII	Four-step phase fitted method of Raptis and Simos (1991)
Method MXXIII	Sixth-order phase fitted method of Simos (1992b)
Method MXXIV	Hybrid phase fitted method of Simos (1993d)
Method MXXV	Hybrid phase fitted method of Simos (1993c)
Method MXXVI	Hybrid fourth-order phase fitted method of Simos (1995b)

The numerical results obtained for the twenty-six methods were compared with the analytic solution of the Woods–Saxon potential. Table 7 shows the maximum absolute error in the computation of all resonances for steplength equal to $h = \frac{1}{2}$. The empty areas indicate that the corresponding maximum absolute errors are larger than 1.

The performance of the different methods is dependent on the choice of the fitting parameter v . For the purpose of obtaining our numerical results it is appropriate to choose v in the way suggested by Ixaru and Rizea (1980). That is, we choose:

$$v = \begin{cases} (-50 - E)^{1/2} & \text{for } x \in [0, 6.5] \\ (-E)^{1/2} & \text{for } x \in [6.5, 15] \end{cases} \quad (184)$$

For a discussion of the reasons for choosing the values 50 and 6.5 and the extent to which the results obtained depend on these values see (Ixaru and Rizea, 1980, p. 25).

10.1.2. Modified Woods–Saxon potential

In Table 8 the maximum absolute error in the computation of all resonances for steplength equal to $h = \frac{1}{2}$ obtained with another potential in Eq. (179) are shown. This potential is

$$V(x) = V_w(x) + \frac{D}{x} \quad (185)$$

where V_w is the Woods–Saxon potential Eq. (179). For the purpose of our numerical experiments we use the

Table 9

Maximum absolute errors, of the eigenvalues calculated by the 26 algorithms MI–MXXVI for the bound-states problem with the Woods–Saxon potential^a

Method	Maximum absolute error MI
MII	5.5×10^{-3}
MIII	6.6×10^{-4}
MIV	8.2×10^{-5}
MV	5.7×10^{-6}
MVI	2.3×10^{-4}
MVII	5.5×10^{-5}
MVIII	3.2×10^{-6}
MIX	5.5×10^{-7}
MX	4.6×10^{-5}
MXI	6.7×10^{-6}
MXII	4.6×10^{-7}
MXIII	7.8×10^{-8}
MXIV	1.1×10^{-8}
MXV	2.3×10^{-9}
MXXVI	4.5×10^{-5}
MXXVII	6.6×10^{-6}
MXXVIII	4.4×10^{-7}
MXXIX	7.1×10^{-8}
MXX	8.0×10^{-9}
MXXI	9.1×10^{-3}
MXXII	8.3×10^{-3}
MXXIII	7.7×10^{-3}
MXXIV	5.2×10^{-3}
MXXV	3.1×10^{-3}
MXXVI	2.2×10^{-4}

^a The empty areas indicate that the maximum absolute error is greater than 1.

same parameters as in (Ixaru and Rizea, 1980), i.e., $D = 20, l = 2$.

Since $V(x)$ is singular at the origin, we use the special strategy of (Ixaru and Rizea, 1980). We start the integration from a point $\epsilon > 0$ and the initial values $y(\epsilon)$ and $y(\epsilon + h)$ for the integration scheme are obtained using a perturbation method (Ixaru, 1984). As in (Ixaru and Rizea, 1980) we use the value $\epsilon = \frac{1}{4}$ for our numerical experiments.

For the purpose of obtaining our numerical results it is appropriate to choose v in the way suggested by Ixaru and Rizea (1980). That is, we choose:

$$v =$$

$$\frac{[V(a_1) + V(\epsilon)]}{2} \quad \text{for } x \in [\epsilon a_1]$$

$$\frac{V(a_1)}{2} \quad \text{for } x \in [a_1, a_2]$$

$V(a_3)$ for $x \in [a_2, a_3]$

$V(15)$ for $x \in [a_3, 15]$.

10.1.3. The bound-states problem

For negative energies we solve the so-called bound-states problem, i.e., with the boundary conditions

$$y(0) = 0, \quad y(x) \sim \exp(-\sqrt{-E}x) \text{ for large } x.$$

In order to solve this problem numerically we use a strategy which has been proposed by Cooley (1961) and has been improved by Blatt (1967). This strategy involves integrating forward from the point $x = 0$, backward from the point $x_b = 15$ and matching up the solution at some internal point in the range of integration. As initial conditions for the backward integration we take (Cash et al., 1990):

$$\begin{aligned} y(x_b) &= \exp(-\sqrt{-E}x_b) \text{ and } y(x_b - h) \\ &= \exp[-\sqrt{-E}(x_b - h)] \end{aligned} \quad (186)$$

where h is the steplength of integration of the numerical method.

The true solutions to the Woods–Saxon bound-states problem were obtained correct to fourteen decimal places using the analytic solution and the numerical results obtained using the above mentioned methods mentioned above were compared to this true solution. In Table 9 we present the maximum absolute error in the computation of all eigenvalues for steplength equal to $h = \frac{1}{2}$. The empty areas indicate that the corresponding absolute errors are larger than 1.

For the purpose of obtaining our numerical results it is appropriate to choose v in the way suggested by Ixaru and Rizea (1980) and given in Section 10.1.1.

10.1.4. Remarks and conclusion

The most accurate fourth-order exponentially fitted method for all problems mentioned above is the method proposed by Simos and Williams (Case VI). The most accurate sixth-order exponentially fitted method for all problems mentioned above is the method proposed by Thomas and Simos (Case V). Finally, the most accurate phase-fitted method for all problems mentioned above is the hybrid method proposed by Simos (1995b).

From above, it is obvious that the crucial concern when solving the Schrödinger equation is that the numerical method should integrate exactly the exponential functions (Eq. (49) with m and p as large as possible.

Table 10

Maximum absolute errors, of the eigenvalues calculated by nine algorithms MI–MIX for the resonance problem with the Woods–Saxon potential^a

Method	Maximum absolute error MI
MII	1.8×10^{-3}
MIII	8.4×10^{-5}
MIV	
MV	4.3×10^{-6}
MVI	9.0×10^{-7}
MVII	6.7×10^{-5}
MVIII	3.8×10^{-6}
MIX	7.5×10^{-6}

^a The empty areas indicate that the maximum absolute error is greater than 1.

10.2. Methods with constant coefficients

We consider the numerical integration of the Schrödinger equation (179) in the well-known case where the potential $V(x)$ is the Woods–Saxon potential (180). As in the previous numerical experiments we take the domain of integration as $0 \leq x \leq 15$. We consider Eq. (179) in a rather large domain of energies, i.e., $E \in [1, 1000]$. The problem we consider here is the so-called *resonance problem*. We note that for the modified Woods–Saxon potential and for the *bound-states problem* the results are similar. The integration is based on the description made in the previous paragraphs.

For comparison purposes we use the following procedures:

Method MI	P-stable method $METH_4(a, b, c)$ of Simos and Raptis (1990).
Method MII	P-stable method of Simos and Mousadis (1994) with phase-lag of order twelve.
Method MIII	P-stable method of Simos (1997d) with phase-lag of order sixteen.
Method MIV	P-stable sixth-order method of Simos (1993e) with phase-lag of order eight.
Method MV	P-stable sixth-order method of Simos (1996c) with phase-lag of order ten.
Method MVI	P-stable sixth-order method of Simos (1997c) with phase-lag of order twenty.
Method MVII	Eighth-order method of Simos (1996d) with phase-lag of order ten.

Method MVIII	Eighth-order method of Simos (1997h) with phase-lag of order eighteen and interval of periodicity equal to $[0, 72.86]$.
Method MIX	Eighth-order method of Simos (1997i) with phase-lag of order fourteen and interval of periodicity equal to $[0, 62.75]$.

The numerical results obtained for the nine methods were compared with the analytic solution of the Woods–Saxon potential. Table 10 shows the maximum absolute error in the computation of all resonances for steplength equal to $h = \frac{1}{2}$. The empty areas indicate that the corresponding maximum absolute errors are larger than 1.

10.2.1. Remarks and conclusion

The most accurate fourth-order P-stable method with constant coefficients is the P-stable method proposed by Simos (1997d) with phase-lag of order sixteen. The most accurate sixth-order method with constant coefficients is the P-stable sixth-order method of Simos (1997e) with phase-lag of order twenty. Finally, the most accurate eighth-order method with constant coefficients is the eighth-order method of Simos (1997h) with phase-lag of order eighteen and interval of periodicity equal to $[0, 72.86]$.

We note that the most accurate methods with constant coefficients for all the problems mentioned above are the eighth-order method of Simos (1997i) with phase-lag of order fourteen and interval of periodicity equal to $[0, 62.75]$ and the eighth-order method of Simos (1997h) with phase-lag of order eighteen and interval of periodicity equal to $[0, 72.86]$.

10.3. Variable-step methods

10.3.1. Error estimation

It is known from the literature (see for example (Shampine et al., 1975) and references therein) that the LTE is based on the algebraic order of the method and that there are many methods for the estimation of the local truncation error (LTE) in the integration of systems of initial-value problems.

The local error estimation algorithm used for our illustration is based on the fact that when we have a method with local error of higher (algebraic or phase-lag or exponential or Bessel and Neumann) order then the approximation of the solution for the problems which have a periodic or oscillating solution is better.

Denoting the solution obtained with higher (algebraic or phase-lag or exponential or Bessel and Neumann) order method as y_{n+1}^H and the solution obtained with lower (algebraic or phase-lag or exponential or Bessel and Neumann) order method as y_{n+1}^L ,

we have the following definition:

Definition 9. We define the local error estimate in the lower order solution y_{n+1}^L by the quantity

$$\text{LTE} = |y_{n+1}^H - y_{n+1}^L|. \quad (187)$$

Under the assumption that when h is sufficiently small, the local error in y_{n+1}^H can be neglected compared with that in y_{n+1}^L .

We assume that the solution y_{n+1}^H is obtained using the a higher (algebraic or phase-lag or exponential or Bessel and Neumann) order method and the solution y_{n+1}^L is obtained using a lower (algebraic or phase-lag or exponential or Bessel and Neumann) order method.

If the local phase-lag error is bounded by acc and the step size of the integration used for the n th step length is h_n , the estimated step size for the $(n+1)$ th step, which will give a local error bounded by acc , must be

$$h_{n+1} = h_n \left(\frac{acc}{\text{LTE}} \right)^{1/q} \quad (188)$$

where q is the order of the local error.

Following (Raptis and Cash, 1985), we have considered all step changes to halving and doubling. Thus, based on the procedure developed by Raptis and Cash (1985), for the LTE, the step control procedure which we use for the Local Error is

If $\text{LTE} < acc$, $h_{n+1} = 2h_n$

If $100 acc > \text{LTE} \geq acc$, $h_{n+1} = h_n$ (189)

If $\text{LTE} \geq 100 acc$,

$$h_{n+1} = \frac{h_n}{2} \text{ and repeat the step.} \quad (190)$$

It is known that the local error estimate is obtained to the lower order solution. This is applied, also, in our case of the local error estimate, i.e. the local error estimate is obtained for y_{n+1}^L . However, if this error estimate is acceptable, i.e. less than the bound acc , we consider the widely used local extrapolation technique. Thus, although we are controlling an estimation of the local error in the lower order solution y_{n+1}^L , we use the higher order solution y_{n+1}^H at each accepted step.

10.4. Coupled differential equations

There are many problems in theoretical physics and chemistry, atomic physics, physical chemistry, quantum

chemistry and chemical physics which can be transformed to the solution of coupled differential equations of the Schrödinger type. The close-coupling differential equations of the Schrödinger type may be written in the form

$$\left[\frac{d^2}{dx^2} + k_i^2 - \frac{l_i(l_i + 1)}{x^2} - V_{ii} \right] y_{ij} = \sum_{m=1}^N V_{im} y_{mj} \quad (191)$$

for $1 \leq i \leq N$ and $m \neq i$.

We have investigated the case in which all channels are open. So we have the following boundary conditions (see for details Allison (1970)):

$$y_{ij} = 0 \text{ at } x = 0 \quad (192)$$

$$y_{ij} \sim k_i x j_{li}(k_i x) \delta_{ij} + \left(\frac{k_i}{k_j} \right)^{1/2} K_{ij} k_i x n_{li}(k_i x) \quad (193)$$

where $j_l(x)$ and $n_l(x)$ are the spherical Bessel and Neumann functions, respectively. We note here that since the methods used for comparison purposes in this review have much larger intervals of periodicity (the property which must have a method to avoid instabilities) than the Numerov's method and other finite difference methods, we can use the present methods to problems involving close channels.

Based on the detailed analysis developed by Allison (1970) and defining a matrix $\mathbf{K}'$ and diagonal matrices $\mathbf{M}$, $\mathbf{N}$ by:

$$\mathbf{K}'_{ij} = \left(\frac{k_i}{k_j} \right)^{1/2} K_{ij}, \quad \mathbf{M}_{ij} = k_i x j_{li}(k_i x) \delta_{ij},$$

$$\mathbf{N}_{ij} = k_i x n_{li}(k_i x) \delta_{ij}$$

we find that the asymptotic condition (193) may be written as:

$$\mathbf{y} \sim \mathbf{M} + \mathbf{N} \mathbf{K}' \quad (194)$$

One of the most well-known methods for the numerical solution of the coupled differential equations arising from the Schrödinger equation is the Iterative Numerov method of Allison (1970).

A real problem in theoretical physics, atomic physics, quantum chemistry and molecular physics which can be transformed to close-coupling differential equations of the Schrödinger type is the rotational excitation of a diatomic molecule by neutral particle impact. Denoting, as by Allison (1970), the entrance channel by the quantum numbers (j, l) , the exit channels by (j', l') , and the total angular momentum by $J = j + l = j' + l'$, we find that

$$\frac{d^2}{dx^2} + k_{j'j}^2 - \frac{l'(l' + 1)}{x^2} y_{j'l'}^{Jl}(x) = \frac{2\mu}{\hbar^2} \sum_{j''} \sum_{l''} \langle j'l'; \quad (195)$$

$$J | V | j''l''; \quad J > y_{j''l''}^{Jl}(x)$$

where

$$k_{j'j} = \frac{2\mu}{\hbar^2} \left[E + \frac{\hbar^2}{2I} \{ j(j+1) - j'(j'+1) \} \right] \quad (196)$$

E is the kinetic energy of the incident particle in the center-of-mass system, I is the moment of inertia of the rotator, and μ is the reduced mass of the system.

Following the analysis of Allison (1970), the potential V may be written as

$$V(x, \hat{\mathbf{k}}_{j'j}, \hat{\mathbf{k}}_{jj}) = V_0(x) P_0(\hat{\mathbf{k}}_{j'j} \hat{\mathbf{k}}_{jj}) + V_2(x) P_2(\hat{\mathbf{k}}_{j'j} \hat{\mathbf{k}}_{jj}), \quad (197)$$

and the coupling matrix element is given by

$$\begin{aligned} \langle j'l'; J | V | j''l''; J \rangle \\ = \delta_{jj''} \delta_{j'l''} V_0(x) + f_2(j'l'; j''l''; J) V_2(x) \end{aligned} \quad (198)$$

where the f_2 coefficients can be obtained from formulas given by Bernstein et al. (1963) and $\hat{\mathbf{k}}_{j'j}$ is a unit vector parallel to the wave vector $\mathbf{k}_{j'j}$ and P_i , $i = 0, 2$ are Legendre polynomials (Bernstein et al., 1963). The boundary conditions may then be written as by Allison (1970)

$$y_{j'l'}^{Jl}(x) = 0 \text{ at } x = 0 \quad (199)$$

$$y_{j'l'}^{Jl}(x) \sim \delta_{jj'} \delta_{ll'} \exp[-i(k_{jj}x - 1/2l\pi)] - \left(\frac{k_i}{k_j} \right)^{1/2} \quad (200)$$

$$S^J(jl; j'l') \exp[i(k_{j'j}x - 1/2l'\pi)]$$

where the scattering $\mathbf{S}$ matrix is related to the $\mathbf{K}$ matrix of (193) by the relation

$$\mathbf{S} = (\mathbf{I} + i\mathbf{K})(\mathbf{I} - i\mathbf{K})^{-1} \quad (201)$$

The calculation of the cross-sections for rotational excitation of molecular hydrogen by impact of various heavy particles requires the existence of the numerical method for step-by-step integration from the initial value to matching points.

In our numerical test we choose the $\mathbf{S}$ matrix which is calculated using the following parameters

$$\frac{2\mu}{\hbar^2} = 1000.0, \quad \frac{\mu}{I} = 2.351, \quad E = 1.1,$$

Table 11

RTC (real time of computation (in seconds)) to calculate $|S|^2$ for the variable-step methods Eqs. (1)–(15)^a

Method	N	hmax	RTC
Iterative Numerov (1970)	4	0.014	3.25
	9	0.014	23.51
	16	0.014	99.15
Variable-step method of Raptis and Cash (1985)	4	0.056	1.55
	9	0.056	8.43
	16	0.056	43.32
Variable-step method of Simos (1996d)	4	0.056	1.05
	9	0.056	5.25
	16	0.056	27.15
Variable-step method of Simos and Williams (1998)	4	0.448	0.24
	9	0.448	0.96
	16	0.448	5.04
Variable-step method of Simos (1997h)	4	0.056	0.35
	9	0.056	1.38
	16	0.056	6.51
Variable-step method of Simos (1997)	4	0.448	0.27
	9	0.448	1.48
	16	0.448	6.31
Variable-step method of Simos (1997e)	4	0.224	0.30
	9	0.224	1.42
	16	0.224	6.43
Variable-step method of Simos (1997d)	4	0.448	0.28
	9	0.448	1.55
	16	0.448	6.37
Variable-step method of Simos and Williams (1997b)	4	0.448	0.18
	9	0.448	0.92
	16	0.224	5.32
Variable-step method of Simos (1993h)	4	0.112	1.43
	9	0.112	8.22
	16	0.112	40.13
Generator of methods of Avdelas and Simos (1996b)	4	0.224	0.83
	9	0.224	6.19
	16	0.224	12.33
Embedded methods of Avdelas and Simos (1996a)	4	0.224	1.03
	9	0.224	6.91
	16	0.224	14.05
Variable-step methods of Simos (1996a)	4	0.112	0.72
	9	0.112	3.20
	16	0.112	14.35
Variable-step methods of Simos (1995b)	4	0.224	0.30
	9	0.224	1.54
	16	0.224	7.42
Variable-step method of Simos (1993g)	4	0.112	0.52
	9	0.112	1.88
	16	0.112	8.04

^a $acc = 10^{-6}$, hmax is the maximum stepsize.

$$V_0(x) = \frac{1}{x^{12}} - 2\frac{1}{x^6}, \quad V_2(x) = 0.2283V_0(x)$$

As is described by Allison (1970), we take $J = 6$ and consider excitation of the rotator from the $j = 0$ state

to levels up to $j' = 2, 4$ and 6 giving sets of *four, nine and sixteen coupled differential equations*, respectively. Following Bernstein (1960) and Allison (1970) the reduction of the interval $(0, \infty)$ to $[0, x_0]$ is obtained. The

wavefunctions are then vanished in this region and consequently the boundary condition (199) may be written as

$$y_{j'l'}^{j,l}(x_0) = 0 \quad (202)$$

For the numerical solution of this problem we have used (1) the well known Iterative Numerov method of Allison (1970), (2) the variable-step method of Raptis and Cash (1985), (3) the variable-step method of Simos (1996d), (4) the variable-step exponentially fitted method of Simos and Williams (1998), (5) the variable-step eighth algebraic order method developed by Simos (1997h), (6) the explicit variable-step method developed by Simos and Tougelidis (1997), (7) the variable-step P-stable sixth algebraic order method developed by Simos (1997c), (8) the embedded P-stable fourth algebraic order method developed by Simos (1977d), (9) the variable-step Bessel and Neumann fitted method developed by Simos and Williams (1997b), (10) the variable-step (of exponential orders eight and ten) P-stable method developed by Simos (1993), (11) the generator of various exponential order P-stable methods developed by Avdelas and Simos (1996b), (12) the embedded sixth algebraic order method developed by Avdelas and Simos (1996a), (13) the exponentially fitted variable-step method developed by Simos (1996a), (14) the variable-step phase-fitted method developed by Simos (1995c) and (15) the variable-step P-stable method developed by Simos (1993g). In Table 11 we present the real time of computation required by the methods mentioned above to calculate the square of the modulus of the S matrix for sets of 4, 9 and 16 coupled differential equations. In Table 11, N indicates the number of equations of the set of coupled differential equations.

10.4.1. Remarks and conclusion

The most efficient variable-step method for the solution of coupled differential equations arising from the Schrödinger equation is the variable-step Bessel and Neumann fitted method of Simos and Williams developed by Simos and Williams (1997b).

Efficient methods are also, the variable-step exponentially fitted method of Simos and Williams (1998), the explicit variable-step method developed by Simos and Tougelidis (1997), the variable-step P-stable sixth algebraic order method developed by Simos (1997e), the embedded P-stable fourth algebraic order method developed by Simos (1997d) and the variable-step phase-fitted method developed by Simos (1995b).

It is obvious the critical properties for efficient variable-step procedures are (1) Bessel and Neumann fitting for scattering problems, (2) exponentially-fitting, (3) P-stability and (4) minimal phase-lag.

10.5. Runge–Kutta methods and methods based on transission line theory

Based on the results presented in the relative papers, the most accurate method is the perturbative method developed by Simos (1995e) and the most efficient variable-step method is the one developed by Simos and Tougelidis (1996b).

Based on the results presented in the relative paper, the most efficient Runge–Kutta method for specific Schrödinger equations is the one developed by Simos and Williams (1996c) with seven stages.

References

- Allison, A.C., 1970. The numerical solution of coupled differential equations arising from the Schrödinger equations. *Journal of Computational Physics* 6, 378–391.
- Allison, A.C., Raptis, A.D., Simos, T.E., 1991. An eighth-order formula for the numerical integration of the one-dimensional Schrodinger equation. *Journal of Computational Physics* 97, 240–248.
- Avdelas, G., Simos, T.E., 1996a. Embedded methods for the numerical solution of the Schrodinger equation. *Computers and Mathematics with Applications* 31, 85–102.
- Avdelas, G., Simos, T.E., 1996b. A generator of high-order embedded P-stable methods for the numerical solution of the Schrodinger equation. *Journal of Computational and Applied Mathematics* 72, 345–358.
- Berstein, R.B., Dalgarno, A., Massey, H., Percival, I.C., 1963. Thermal scattering of atoms by homonuclear diatomic molecules. *Proceedings of the Royal Society of London-Series A* 274, 427–442.
- Berstrein, R.B., 1960. Quantum mechanical (phase shift) analysis of differential elastic scattering of molecular beams. *Journal of Chemical Physics* 33, 795–804.
- Blatt, J.M., 1967. Practical points concerning the solution of the Schrödinger equation. *Journal of Computational Physics* 1, 382–396.
- Cash, J.R., Raptis, A.D., Simos, T.E., 1990. A sixth-order exponentially fitted method for the numerical solution of the radial Schrodinger equation. *Journal of Computational Physics* 91, 413–423.
- Coleman, J.P., 1989. Numerical methods for $y''=f(x,y)$ via rational approximation for the cosine. *IMA Journal of Numerical Analysis* 9, 145–165.
- Cooley, J.W., 1961. An improved eigenvalue corrector formula for solving Schrödinger's equation for central fields. *Mathematics of Computation* 15, 363–374.
- Fack, V., Vanden Berghe, G., 1985. A finite difference approach for the calculation of perturbed oscillator energies. *J. Phys. A: Math. Gen* 18, 3355–3363.
- Fack, V., Vanden Berghe, G., 1987. (Extended) Numerov method for computing eigenvalues of specific Schrödinger equations. *J. Phys. A: Math. Gen.* 20, 4153–4160.
- Hairer, E., Norsett, S.P., Wanner, G., 1987. *Solving Ordinary Differential Equations I*. Springer–Verlag, Berlin.

- Henrici, P., 1962. *Discrete Variable Methods in Ordinary Differential Equations*. Wiley, New York.
- Herzberg, G., 1950. *Spectra of Diatomic Molecules*. Van Nostrand, Toronto.
- Ixaru, L.Gr., Rizea, M., 1980. A Numerov-like scheme for the numerical solution of the Schrödinger equation in the deep continuum spectrum of energies. *Computer Physics Communications* 19, 23–27.
- Ixaru, L.Gr., 1984. *Numerical Methods for Differential Equations and Applications*. Reidel, Dordrecht, Boston, Lancaster.
- Johnson, B.R., 1978. The renormalized Numerov method applied to calculating bound states of the coupled-channel Schrödinger equation. *Journal of Chemical Physics* 69, 4678–4688.
- Lambert, J.D., Watson, I.A., 1976. Symmetric multistep methods for periodic initial value problems. *Journal of Institute of Mathematics and its Applications* 18, 189–202.
- Lyche, T., 1972. Chebyshevian multistep methods for ordinary differential equations. *Numerische Mathematik* 10, 65–75.
- Papageorgiou, C.D., Raptis, A.D., 1987. A method for the solution of the Schrödinger equation. *Computer Physics Communications* 43, 325–328.
- Papageorgiou, C.D., Raptis, A.D., Simos, T.E., 1990a. A method for computing phase shifts for scattering. *Journal of Computational and Applied Mathematics* 29, 61–67.
- Papageorgiou, C.D., Raptis, A.D., Simos, T.E., 1990b. An algorithm for the solution of the eigenvalue Schrödinger equation. *Journal of Computational Physics* 88, 477–483.
- Press, W.H., Teukolsky, S.A., Vetterling, W.T., Flannery, B.P., 1992. *Numerical Recipes in C. The Art of Scientific Computing*, Cambridge University Press.
- Raptis, A.D., 1981. On the numerical solution of the Schrödinger equation. *Computer Physics Communications* 24, 1–4.
- Raptis, A.D., 1982. Two-step methods for the numerical solution of the Schrödinger equation. *Computing* 28, 373–378.
- Raptis, A.D., 1983. Exponentially fitted solutions of the eigenvalue Schrödinger equation with automatic error control. *Computer Physics Communications* 28, 427–431.
- Raptis, A.D., 1984. Exponential multistep methods for ordinary differential equations. *Bulletin of the Greek Mathematical Society* 25, 113–126.
- Raptis, A.D., Allison, A.C., 1978. Exponential - fitting methods for the numerical solution of the Schrödinger equation. *Computer Physics Communications* 14, 1–5.
- Raptis, A.D., Cash, J.R., 1985. A variable step method for the numerical integration of the one-dimensional Schrödinger equation. *Computer Physics Communications* 36, 113–119.
- Raptis, A.D., Cash, J.R., 1987. Exponential and Bessel fitting methods for the numerical solution of the Schrödinger equation. *Computer Physics Communications* 44, 95–103.
- Raptis, A.D., Simos, T.E., 1991. A four-step phase-fitted method for the numerical integration of second-order initial-value problem. *BIT* 31, 160–168.
- Simos T.E. Numerical solution of ordinary differential equations with periodical solution. Doctoral Dissertation. National Technical University of Athens, 1990a.
- Simos, T.E., 1990b. A four-step method for the numerical solution of the Schrodinger equation. *Journal of Computational and Applied Mathematics* 30, 251–255.
- Simos, T.E., 1991a. Some new four-step exponential fitting methods for the numerical solution of the radial Schrodinger equation. *IMA Journal of Numerical Analysis* 11, 347–356.
- Simos, T.E., 1991b. A two-step method with phase-lag of order infinity for the numerical integration of second-order periodic initial- value problems. *International Journal of Computer Mathematics* 39, 135–140.
- Simos, T.E., 1991c. A Numerov-type method for the numerical solution of the radial Schrodinger equation. *Applied Numerical Mathematics* 7, 201–206.
- Simos, T.E., 1992a. Exponential fitted methods for the numerical integration of the Schrodinger equation. *Computer Physics Communications* 71, 32–38.
- Simos, T.E., 1992b. Two-step almost P-stable complete in phase methods for the numerical integration of second-order periodic initial-value problems. *International Journal of Computer Mathematics* 46, 77–85.
- Simos, T.E., 1992c. An explicit almost P-stable two-step method with phase-lag of order infinity for the numerical integration of second-order periodic initial-value problems. *Applied Mathematics and Computation* 49, 261–268.
- Simos, T.E., 1992d. Explicit two-step methods with minimal phase-lag for the numerical integration of special second-order initial value problems and their application to the one-dimensional Schrodinger equation. *Journal of Computational and Applied Mathematics* 39, 89–94.
- Simos, T.E., 1993a. Runge–Kutta interpolants with minimal phase-lag. *Computers and Mathematics with Applications* 26, 43–49.
- Simos, T.E., 1993b. Error analysis of exponential-fitted methods for the numerical solution of the one-dimensional Schrödinger equation. *Physics Letters A* 177, 345–350.
- Simos, T.E., 1993c. A family of two-step almost P-stable methods with phase-lag of order infinity for the numerical integration of second-order periodic initial-value problems. *Japan Journal of Industrial and Applied Mathematics* 10, 289–297.
- Simos, T.E., 1993d. A predictor–corrector phase-fitted method for $y'' = f(x,y)$. *Mathematics and Computers in Simulation* 35, 153–159.
- Simos, T.E., 1993e. High-order methods with minimal phase-lag for the numerical integration of the special second-order initial-value problem and their application to the one-dimensional Schrodinger equation. *Computer Physics Communications* 74, 63–66.
- Simos, T.E., 1993f. A new variable step method for the numerical integration of special second-order initial value problems and their application to the one-dimensional Schrodinger equation. *Applied Mathematics Letters* 6, 67–73.
- Simos, T.E., 1993g. A variable-step procedure for the numerical integration of the one-dimensional Schrodinger equation. *Computers in Physics* 7, 460–464.
- Simos, T.E., 1993h. New variable-step procedure for the numerical integration of the one-dimensional Schrodinger equation. *Journal of Computational Physics* 108, 175–179.
- Simos, T.E., 1994a. An explicit four-step phase-fitted method

- for the numerical integration of second-order periodic initial-value problems. *Journal of Computational and Applied Mathematics* 55, 125–133.
- Simos, T.E., 1994b. Some new variable-step methods with minimal phase-lag for the numerical integration of special 2nd-order initial value problems. *Applied Mathematics and Computation* 64, 65–72.
- Simos, T.E., 1995a. A family of four-step exponentially fitted predictor–corrector methods for the numerical integration of the Schrödinger equation. *Journal of Computational and Applied Mathematics* 58, 337–344.
- Simos, T.E., 1995b. Predictor–corrector phase-fitted methods for $y'' = f(x, y)$ and an application to the Schrödinger equation. *International Journal of Quantum Chemistry* 53, 473–483.
- Simos, T.E., 1995c. Some low-order two-step almost P-stable methods with phase-lag of order infinity for the numerical integration of the radial Schrödinger equation. *International Journal of Modern Physics A* 10, 2431–2438.
- Simos, T.E., 1995d. An explicit high order predictor–corrector method for periodic initial-value problems. *Mathematical Models and Methods in Applied Sciences* 5, 159–166.
- Simos, T.E., 1995e. A new method for computing eigenvalues of the radial Schrödinger equation. *Chemical Physics Letters* 235, 321–326.
- Simos, T.E., 1996a. Exponential fitted methods for the numerical solution of the Schrödinger equation. *Journal of Computational Mathematics* 14, 120–134.
- Simos, T.E., 1996b. A new Numerov-type method for computing eigenvalues and resonances of the radial Schrödinger equation. *International Journal of Modern Physics C* 7, 33–41.
- Simos, T.E., 1996c. Accurate computations for the elastic scattering phase-shift problem. *Computers and Chemistry* 21, 125–128.
- Simos, T.E., 1996d. An eighth-order method with minimal phase-lag for accurate computations for the elastic scattering phase-shift problem. *International Journal of Modern Physics C* 7, 825–835.
- Simos, T.E., 1997a. An accurate methods for the numerical solution of the Schrödinger equation. *Modern Physics Letters A* 12, 1891–1900.
- Simos, T.E., 1997b. An exponentially fitted method for the numerical solution of the Schrödinger equation. *Journal of Chemical Information and Computer Sciences* 37, 343–348.
- Simos, T.E., 1997c. A simple accurate method for the numerical solution of the Schrödinger equation. *Helvetica Physica Acta* 70, 781–792.
- Simos, T.E., 1997d. New Numerov-type methods for computing eigenvalues, resonances and phase shifts of the radial Schrödinger equation. *International Journal of Quantum Chemistry* 62, 467–475.
- Simos, T.E., 1997e. New P-stable high-order methods with minimal phase-lag for the numerical integration of the radial Schrödinger equation. *Physica Scripta* 55, 644–650.
- Simos, T.E., 1997f. Eighth-order methods for elastic scattering phase-shifts. *International Journal of Theoretical Physics* 36, 663–672.
- Simos, T.E., 1997g. An explicit eighth-order method with minimal phase-lag for the numerical solution of the Schrödinger equation. *Computational Material Science* 8, 317–326.
- Simos, T.E., 1997h. Eighth-order methods for accurate computations for the Schrödinger equation. *Computer Physics Communications* 105, 127–138.
- Simos, T.E., 1997i. Eighth-order methods with minimal phase-lag for accurate computations for the elastic scattering phase-shift problem. *Journal of Mathematical Chemistry* 21, 359–372.
- Simos, T.E., 1998. An accurate exponentially fitted four-step method for the numerical solution of the radial Schrödinger equation. *Molecular Simulation* 20, 285–301.
- Simos, T.E., Mitsou, G.V., 1994. A family of four-step exponential fitted methods for the numerical integration of the radial Schrödinger equation. *Computers and Mathematics with Applications* 28, 41–50.
- Simos, T.E., Mousadis, G., 1994. Some new Numerov-type methods with minimal phase-lag for the numerical integration of the radial Schrödinger equation. *Molecular Physics* 83, 1145–1153.
- Simos, T.E., Mousadis, G., 1995. A two-step method for the numerical solution of the radial Schrödinger equation. *Computers and Mathematics with Applications* 29, 31–37.
- Simos, T.E., Raptis, A.D., 1990. Numerov-type methods with minimal phase-lag for the numerical integration of the one-dimensional Schrödinger equation. *Computing* 45, 175–181.
- Simos, T.E., Raptis, A.D., 1992. A fourth-order Bessel fitting method for the numerical solution of the Schrödinger equation. *Journal of Computational and Applied Mathematics* 43, 313–322.
- Shampine, L.F., Watts, H.A., Davenport, S.M., 1975. Solving non-stiff ordinary differential equations-the state of the art. *SIAM Review* 18, 376–411.
- Simos, T.E., Tougelidis, G., 1996a. A Numerov-type method for computing eigenvalues and resonances of the radial Schrödinger equation. *Computers and Chemistry* 20, 397–401.
- Simos, T.E., Tougelidis, G., 1996b. A variable-step algorithm for computing eigenvalues of the radial Schrödinger equation. *International Journal of Quantum Chemistry* 59, 477–485.
- Simos, T.E., Tougelidis, G., 1997. An explicit eighth-order method with minimal phase-lag for accurate computations of eigenvalues, resonances and phase shifts. *Computers and Chemistry* 21, 327–334.
- Simos, T.E., Williams, P.S., 1996c. Some modified Runge–Kutta methods for the numerical solution of some specific Schrödinger equations and related problems. *International Journal of Modern Physics A* 11, 4731–4744.
- Simos, T.E., Williams, P.S., 1997a. A family of Numerov-type exponentially fitted methods for the numerical integration of the Schrödinger equation. *Computers and Chemistry* 21, 403–417.
- Simos, T.E., Williams, P.S., 1997b. Bessel and Neumann fitted methods for the numerical solution of the radial Schrödinger equation. *Computers and Chemistry* 21, 175–179.
- Simos, T.E., Williams, P.S., 1997c. A finite-difference method for the numerical solution of the Schrödinger equation. *J. Comput. Appl. Math* 79, 189–205.

- Simons, T.E., Williams, P.S., 1998. Computer algebra programmes for the construction of a family of Numerov-type exponentially fitted methods for the numerical solution of the Schrodinger equation. *Computers and Chemistry* 22, 185–218.
- Thomas, R.M., 1984. Phase properties of high order, almost P-stable formulae. *BIT* 24, 225–238.
- Thomas, R.M., Simos, T.E., 1997. A family of hybrid exponentially fitted predictor–corrector methods for the numerical solution of the radial Schrodinger equation. *Journal of Computational and Applied Mathematics* 87, 215–226.
- Thomas, R.M., Simos, T.E., Mitsou, G.V., 1996. A family of Numerov-type exponentially fitted predictor–corrector methods for the numerical integration of the radial Schrodinger equation. *Journal of Computational and Applied Mathematics* 67, 255–270.
- van der Houwen, P.J., Sommeijer, B.P., 1987. Explicit Runge–Kutta(–Nyström) methods with reduced phase errors for computing oscillating solutions. *SIAM Journal of Numerical Analysis* 24, 595–617.